LPile Plus Student Edition
Electronic Documentation

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Introduction to LPILE Plus Student Edition

LPILE Plus Student Edition provides an interactive way to enter input data, perform an analysis of a laterally loaded pile, and view the output from the analysis.

The main menu for LPile Plus Student Edition is displayed at the top of the window. It has seven choices for sub-menus: File, Data, Options, Computation, Graphics, Window, and Help. When a graph is displayed, an eighth option, Plot, is displayed

The File menu allows the user to access a pull down menu for file operations such as opening an existing file, creating a new data file, and for exiting the program.

The Data menu allows the user to enter required data for computing the response of the pile and soil, such as distributions of deflection, bending moment, shear, and soil resistance as a function of length along the pile.

The Options menu allows the user to enter data for controlling the computation and output options. You can use the default settings if no change is needed.

The Computation menu is selected to execute the program using the current input data. The user can also view the text data from the input and output files.

The Graphics menu is selected to observe the different graphical representations of the program results contained in the output file. When a graph is displayed on the screen, the Plot menu is used to control output of the currently displayed graph.

The Window menu is used for arranging open windows on the screen.

The Help menu provides access to this help system.

Limitations of LPile Plus Student Edition

LPile Plus Student Edition is limited to the analysis of problems with the following restrictions:

Piles are limited in size piles. The maximum size of pile is 24 inches in diameter and 75 feet in length.

Soil profiles are limited to a single soil layer.

Distributed lateral loading along the side of the pile is not allowed.

Piles are limited to elastic pile sections only. Piles with nonlinear moment-curvature behavior, such as drilled shafts and prestressed concrete piles are not permitted.

In addition, the data files produced by LPile Plus Student Edition are not compatible with the professional version of LPile Plus for Windows.

Pull-down Menus

The pull-down menus may be accessed by

Pressing the Alt key and the underlined letter of the menu option or

By positioning the mouse track over the menu item and clicking the *left* mouse button.

You can then select an item on the drop-down menu by either typing the underlined letter for the menu selection (don't press Alt this time), or by clicking on the selection with the mouse.

Once you have opened one of the menus, you can use the right arrow and left arrow keys to move between them. Some of the menu selections may not be available at a given time; these will be shown in gray lettering.

Keyboard Short Cuts

The following short-cut menu accelerators (or "Hot Keys") can be used:

F1 calls up this Help system.

Alt-F4 (the Alt and F4 keys pressed simultaneously) can be used to exit LPPEDIT from any location in the program. Note that all data will be lost when Alt-F4 is pressed.

Control-F4 closes the currently active LPPEDIT table or dialog window.

Shift-F5 "cascades" all the un-iconized windows you have open. It arranges them neatly one on top of the other, each a little lower and to the left of the one before it.

Control-F6 takes you to the next window.

When you are editing a table, you get an additional menu, the *Edit* menu. These accelerators are available for its options:

Shift-End: *Append Row.* Adds a new row at the end of the table.

Shift-Ins: *Insert Row*: Adds a new row at the at the current cursor location within the table.

Shift-Del: *Delete Row*. Deletes the row at the current cursor location within the table.

In-Line Equation Evaluation

Input cells that require numeric data may accept entries of mathematical expressions in addition to simple numeric entries. Entering a mathematical expression works similarly to normal numeric data. The user type the expression that represents the data and presses the Enter key to calculate the entered expression and to display the numeric result in the entry cell.

The table below shows the list of supported operations and constants. The order of operations follows the order in the table. Note that implicit multiplication (i.e. 2(4+6)) is not supported (instead use 2*(4+6) for the previous example).

Scientific notation by entering a lower-case e (i.e. 29e6 or 7.0e-3) is supported. After an expression is evaluated, very large or very small numbers will be displayed using scientific notation.

Symbol	Description
( )	parenthesis (may be nested)
^	exponentiation
*	multiplication
/	division
+	addition
–	subtraction
–	negation (same as subtraction)

Trigonometric function operations in radians are supported. The functions supported are listed in the table below.

Function	Description	Type of Argument	Type of Result
Cos( )	cosine	radians	real number –1 to +1
Sin( )	sine	radians	real number –1 to +1
Tan( )	tangent	radians	real number
Arccos( )	inverse cosine	real number –1 to +1	radians
Arcsin( )	inverse sine	real number –1 to +1	radians
Arctan( )	inverse tangent	real number	radians $-\pi/2$ to $+\pi/2$

Two numerical constants supported by LPile Plus Student Edition are π (entered as pi or PI) and e (entered as upper-case E). Implicit multiplications using constants are not supported. The user should exercise caution when using E to make sure that the computed result is not confused with an expression using scientific notation.

About Ensoft, Inc.

Ensoft, Inc. was founded in 1985 by Professor Lymon C. Reese to develop and market state-of-the-art software for design and analysis of deep foundations.

Ensoft, Inc. is located in Austin, Texas USA.

LPILE Plus Student Edition for Windows was developed and programmed by Dr. William M. Isenhower, Ph.D., P.E.

Technical Support and Program Updates

Technical support is not available to student users of LPile Plus Student Edition.

Technical support is available only to faculty who are currently using LPile Plus Student Edition in classes that have adopted *Analysis and Design of Shallow and Deep Foundations* by Reese, Isenhower, and Wang for the course textbook.

The URL's for the Ensoft web site is http://www.ensoftinc.com/. Users can use Windows Explorer (with view file details turned on) to check the date of their current programs before downloading from the website. Please check the web site or call Ensoft, Inc. for information about latest program dates or for additional information.

Please note that Ensoft maintains only current versions of it's programs for changes in

the Microsoft Windows operating system. Occasionally, Microsoft introduces changes that require changes in a program to run under the new operating system.

Contact Ensoft, Inc. at (512) 244-6464 for further information about program updates or other programs.

Step by Step Instructions for New Users

The following are instructions for new users to LPile Plus Student Edition. These instructions were written assuming that the user has access to the User's Guide and Technical Manual for the program. These instructions also assume that the user is a professional civil engineer who has competent knowledge of soil mechanics, foundation engineering, and structural engineering.

Prior to starting the program, the design engineer should collect the pertinent information needed for design of the laterally loaded foundations. This information should include the both the geotechnical investigation report, the site plan, and the structural loading and performance requirements for the structural foundation. The user should identify what the requirements for the foundation design are and how these requirements are to be evaluated in the design computations. As with any engineering design, any assumptions made and used in design computations must be written down and included in the design file kept by the engineer-of-record.

The engineer begins the analysis by LPile Plus Student Edition by starting the program using the Start Menu or other program short cut. The following are the common steps taken for a basic analysis:

Step 1: Select the Units of computation .

Step 2: Select the Analysis type.

Step 3: Enter the title.

Step 4: Enter the Pile Properties.

Step 5: Enter the Loading Type.

Step 6: Enter the Pile-Head Boundary Conditions and Loading.

Step 7: Soil Layers and Types.

Step 8: Run Analysis. If any errors are detected by the program, the user should follow the directions in the message boxes to correct the input data.

Step 9: Evaluation by the engineer of the computed results to evaluate whether the foundation meets the performance requirements for the structure. The graphs available under the Graphics Menu are provided to aid the designing engineer in this task.

The optimum length of a given pile size can be determined using the Generate Pile Length vs. Top Deflection option.

Output of *p-y* curves at selected depths is available by checking the option for Print p-y Curves. Should very short foundations need to be analyzed, the user should consider using the option for Include Shear Resistance at Pile Tip.

The designing engineer will typically find that range of combinations of pile diameters and lengths will satisfy the functional requirements of the foundation. The designing engineer can compute the cost of each diameter and length combination to determine the most economical option.

Comparison of Soil Types for *p-y* Curves and Axial Load-Transfer

The names for soil types used for axial capacity computations for drilled shafts using the methods presented by O'Neill and Reese (1999) differ from those used for *p-y* curves. The following information is provided to assist users in matching the soil type models for axial and lateral loading of drilled shafts.

Soil Type Names for Axial Capacity Soil Models

Cohesionless Soils have SPT N-values of 50 blows/ft or less (50 blows/0.3 m).

Cohesionless Intermediate Geomaterials have SPT N-values over 50 blows/ft.

Cohesive Soils have undrained shear strength values of less than 37 psi (0.5 MPa). Note that the undrained shear strength is one-half of the uniaxial compressive strength for the same material.

Cohesive Intermediate Geomaterials have uniaxial compressive strengths ranging from 74 psi to 750 psi (0.5 to 5.0 MPa). Note that the uniaxial compressive strength is twice the value of the undrained shear strength.

Rock refers to cohesive materials that have uniaxial compressive strengths higher than 750 psi (5.0 MPa).

Recommended Match-up of Axial Capacity Soil Models with *p-y* Curve Models

Cohesionless Soil: Use either the Sand or API Sand *p-y* curve model.

Cohesionless Intermediate Geomaterial: If uncemented, use either the Sand or API Sand *p-y* curve model. If cemented, use the Weak Rock *p-y* curve model.

Cohesive Soils: If the undrained strength is 1,000 psf or less use the Soft Clay by Matlock (1970). If the undrained strength is over 1,000 psf use one of the Stiff Clay models by Reese et al. (1974). The choice between the models for stiff clay with or without free water will depend on whether free water may enter a gap formed around the top of the pile during cyclic loading.

Cohesive Intermediate Geomaterial: Use the Weak Rock p-y curve model

Rock: In most cases, use the vuggy limestone model. However, in many cases, the rock is stronger than the concrete used in the drilled shaft. For lateral loading considerations in these cases, the rock socket only needs to be longer than the development length for the rebar in the drilled shaft, but the shear steel in the drilled

shaft should be properly sized to carry the high shear forces developed in the shaft near the top of rock. In cases where the rock is strong, the drilled shaft socket will fail structurally at the top of socket before the rock will fail.

Structural Design Issues

Structural Resistance Factors

LPile Plus Student Edition does not include resistance factors in the computations performed by the program. The reason for this is that both the lateral load-transfer from the pile or drilled shaft to the soil and the bending stiffness versus bending moment relationship for the pile or drilled shaft are nonlinear. In order to make accurate computations of lateral deflection and pile-head rotation, it is necessary to use the nonlinear relationships. This type of computation does not permit the inclusion of structural resistance factors.

However, it is necessary for the designing engineer to examine the magnitude of the moments developed in the piles and drilled shafts and to compare these values to the factored moment capacities for the foundation being analyzed.

At the time of writing, the following structural resistance factors are in wide use in the United States:

For bridges for both piles and drilled shafts, the AASHTO resistance factor is 0.75.

For non-bridge structures, ACI does not make any specific recommendations for drilled shafts. However, ACI 318 stipulates the use of 0.75 for columns with spiral transverse reinforcement and 0.70 for tied transverse reinforcement. The requirements for UBC are the same as ACI.

Users outside of the United States should consult the requirements of their local building code for required values for structural resistance factors.

Design Load Combinations

The design loading combinations used by the designer will vary depending on the design code that governs the structure and foundations. There is no perfect agreement on loading combinations between the various design codes. The designing engineer must determine what code is in effect.

In the United States, the commonly-used codes are the Uniform Building Code, the International Building Code (Section 1605), ACI 318 Building Code Requirements for Reinforced Concrete (Section 9.2), AASHTO LRFD Highway Bridge Design Specifications, and the National Fire Protection Association design standards, and other locally dictated design codes.

Tool Bar

The toolbar provides quick access to many of the frequently-used features of LPile. Each speed button on the toolbar provides a bubble help description of the speed buttons function. The bubble help hint is displayed by positioning the mouse cursor over a button. A brief description of each button is shown below.

File New

Open File

Display File Save As dialog

Project Information

Pile Properties

Loading Type

Pile-Head Loading

Display Soil Layers dialog

Analyze pile response

View output file

Display pile and soil profile

Graph *p-y* curves

Graph lateral pile deflection versus depth

Graph bending moment versus depth

Graph shear force versus depth

Graph pile top deflection versus pile embedment length

File Menu

The File Menu controls the commands for file operations. The File Menu options are:

File Menu - New to create a new empty data file

File Menu - Read to open an existing data file

File Menu - Save to save the current data under the current filename

File Menu - Save As to save the current data under a new filename

File Menu - Quit to end the program

File Menu - New

This command closes any open windows and their associated data file, and allows you to start entering data for a new file with a default name of "NewPile". This file is not actually created until you issue the Save command or run an analysis.

If you have unsaved edits in a currently open file, you will be asked if you wish to save or abandon these edits or cancel this command.

This command can also be invoked using the button on the Tool Bar.

File Menu - Open

This command allows the user to enter the name of an existing data to be read for editing. This option will replace any data in the editor with the new data obtained from the input file. If you have unsaved edits in a currently open file, you will be asked if you wish to save or abandon these edits or cancel this command.

This command can also be invoked using the button on the Tool Bar.

File Menu - Save

This command saves any edits to the existing data to the current filename.

Please note that LPile Plus Student Edition will automatically save the current data before performing an analysis. If the user wishes to save several data files, the user must save the files under different filenames using the File Menu - Save As command.

File Menu - Save As

This command writes the data to a new file or overwrites any specified existing file.

The user enters the file name to be saved in a file dialog box.

Immediately below the edit line for the file name is a drop-down list box in which you can scroll the names of file types in a directory.

This command can also be invoked using the button on the Tool Bar.

File Menu - Exit

This command closes the current file and terminates program. If any editing has taken place since the last file save, the user will be asked to either save or abandon the edits or to cancel this command.

Options Menu

The Options menu provides access to the commands for the selection of units of computation, analysis type, various execution options, optional output, and program control options.

Engineering Units Used in Computations

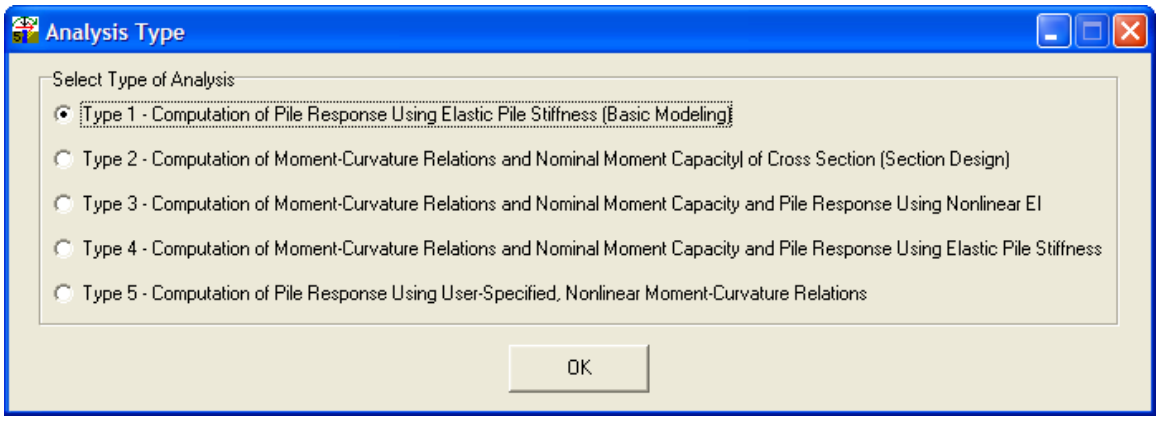
The Units command under the Options menu allows the user to select either US Customary Units, using pounds and inches) or SI Units (using kilonewtons and meters) or any other consistent set of units of force and length. US Customary units are enabled, as a default, for all new runs. The user may change the system of units as many times as desired and values that were previously inputted will be automatically converted by the program between US Customary and SI units and vice versa.

Throughout this help system, L will signify the selected unit of Length and F the selected unit of Force.

Make sure that all input values are entered in consistent units

Analysis Type

LPile Student Edition is restricted to Analysis Type 1. The following information is provided for information about the full professional version.



The Analysis Type dialog box allows the user to choose the type of analytical run that is desired to be performed, the user may select to run one of the four choices described below.

Type 1 - Computations of Pile Response with User-Specified, Constant *EI*

This analysis type performs the analysis of a laterally-loaded pile using flexural stiffnesses computed from the values of moment of inertia (*I*) and modulus of elasticity (*E*) that are input in the dialog box for Pile Properties under the Data menu.

Type 2 - Computations of Ultimate Bending Moment of Cross Section (Section Design)

This analysis type performs only an analysis of the pile's cross section. This type of analysis is used to compute the ultimate bending moment and the nonlinear variation of

flexural stiffness with applied moment. Selection of this option activates the commands at the bottom of the Data pull-down menu. These computations require that information on cross-sectional data be entered in the dialog boxes activated by these commands (Cross-Sectional Shape, Cross-Sectional Dimensions, Cross-Sectional Rebar Arrangement, Cross-Sectional Material Properties and Axial Loads for Cross-Section Analysis).

Type 3 - Computations of Ultimate Bending Moment and Pile Response using Nonlinear *EI*

Selection of this analysis type the program first performs an analysis of the cross section to obtain the ultimate bending moment and the variation of flexural stiffness with applied bending moment. In the second part of the computations, the program performs an analysis of a laterally-loaded pile using the nonlinear flexural stiffnesses computed in the cross-section analysis (using the internally-generated values obtained in the first part of computations).

One special type of analysis performed with the Type 3 analysis is a "push-over analysis. The purpose of performing a push-over analysis is to determine the level of loading and/or deflection that results in the development of a plastic hinge in the pile. In a push-over analysis, the displacement and moment pile-head boundary condition is specified and a number of increasing pile-head displacements are specified until a plastic hinge is developed in the pile. When performing a push-over analysis in which a plastic hinge develops, it is usually necessary to set the maximum iterations to 300.

Type 4 - Computations of Ultimate Bending Moment and Pile Response with User-Specified *EI*

With selection of this analysis type, the program first performs an analysis of the cross section to obtain the ultimate bending moment and the variation of flexural stiffness with applied bending moment. In the second part of the computations, the program performs an analysis of a laterally-loaded pile using flexural stiffnesses computed from the values of moment of inertia (*I*) and modulus of elasticity (*E*) that are inputted in the section of Pile Properties under the Data menu.

Type 5 – Computation of Pile Response Using User-Specified, Nonlinear *EI*
This analysis type allows the user to input nonlinear *EI* vs. moment data for up to five sections along the foundation. A typical application of this option is the analysis of rock-socketed piers, in which the section above the rock is different in size and configuration from the rock socket.

Printing Depths of *p-y* Curves

LPILE Plus can calculate and print *p-y* curves at selected depths for verification purposes. The user may select up to ten different printout depths. These *p-y* curves are calculated expressly for the specified depths. In the LPILE Plus computations of pile response, *p-y* curves are calculated at every nodal point along the length of the pile.

The specified depths must be between the ground surface or the pile head, whichever is lower in elevation, and the pile tip. If inadmissible depths are specified, an error message will be displayed by the program.

User-specified *p-y* curves cannot be output using this option. However, the program always echo prints the input data.

Enter the x-coordinates (L) at which internally generated curves are to be printed. Internally-generated *p-y* curves may be computed for selected depths and printed for verification purposes. In the analysis of pile response, a separate *p-y* curve is calculated at every pile station. Therefore, the number of *p-y* curves printed will have no effect on the solution.

Include *p-y* Modification Factors

The selection of this option indicates that the user will input modification factors for *p-y* curves. A check mark on this option enables the Modification Factors for *p-y* Curves option under the Data menu.

Include Shear Resistance at Pile Tip

The selection of this option includes the use of an externally-specified curve to account for the shear resistance at the pile tip. A check mark on this option enables the Shear-Resistance Curve at Pile Tip option under the Data menu.

Generate Top Deflection vs. Pile Length

If a check mark is placed on this option, the program automatically reduces the specified pile length and makes consecutive analytical runs for up to ten different pile lengths. The additional pile lengths are internally generated based on the inputted value of pile length (in Pile Properties option under the Data menu). The inputted length is reduced in nine equal decrements to about 50% of the initial value.

In the output file and graphics, the user is provided with a plot of pile length versus top deflection that is helpful for determining the critical pile length. The output file and plot may contain less than ten different curve points (pile lengths) if excessive pile-head deflections are found in any particular lengths. If no results are provided, it is possible that the user is already too close to the critical length and any further decrement causes excessive deflections and no valid output. In those cases, it is suggested that the user increases the original pile length so that the program can provide several curve points on the plot of pile length versus top deflection. If the pile length is increased, the user must remember to specify soil parameters that reach the tip of the new pile length. It is recommended to reduce the specified pile length if a horizontal line is obtained in the plot of pile length versus top deflection.

Users are strongly recommended not to select this option on initial runs until being assured to have a working analytical model. This option considerably increases computation time.

Generate Foundation Stiffness

If a check mark is placed on this option, the program automatically selects up to ten additional lateral loads and moments based on the initial values specified by the user. Multiple analytical runs are automatically produced to determine the nonlinear variation of foundation stiffness.

The program selects the additional loads based on the initial values specified by the user in the Boundary Conditions and Loading window screen under the Data menu. This

option can only be used if the user selects to input Shear [F] and Moment [F-L] as boundary conditions. The program automatically reduces the user-specified values of lateral load and moment in ten decrements to zero. Computations are performed for all twenty runs to form the curves of foundation stiffness for a range of loading. Higher values of lateral load and/or moment should be inputted by the user if curves of foundation stiffness for a wider range of loading are desired.

In the output file and graphics, the user is provided with plots of four values of a matrix that can be used to represent the foundation stiffness in the superstructure analysis for a certain range of loading. The axial component of the foundation-stiffness matrix must be calculated separately perhaps with the help of computer programs designed to find the axial capacity and short-term settlements of drilled shafts or driven piles.

Users are strongly recommended not to select this option on initial runs until being assured to have a working analytical model. This option considerably increases computation time.

Only Print Summary Tables

By default, values of lateral deflection, moment, shear force, and soil reaction are output for the full length of the pile being analyzed.

A check mark on this option indicates that the program will include a printout of a summary table in the output file. With this option selected, the output file will contain an echo print of all the inputted information and a table of results that will only include deflection, maximum bending moment, and maximum shear force at the pile head.

Soil Movement

The user may select the Soil Movement option to include the effects of a displaced soil profile on pile response. This option will enable the Soil Movement command on the Data pull-down menu to allow the user to enter the soil movement profile versus depth.

Printing Increment

This option opens a dialog box in which the user can specify the spacing of finite-difference nodes where results will be printed in the table of output.

As a default, results are printed at every nodal point along the pile length. This option is disabled if the user has a check mark on the Only Print Summary Tables option under the Options menu. A value of 1 prints the values at every node; 2 prints values at every other nodes; etc.

Control Options

The Control Options dialog box allows you to edit the controls used for the LPILE Plus analysis. These controls are:

Iteration Limit

The maximum number of iterations performed by the program for the pile solution can be set by the user. Many problems will converge in under 100 iterations unless a plastic

hinge is being developed in an analysis using nonlinear *EI*. If a push-over analysis is being performed using the displacement and moment pile-head boundary condition, the iterations limit should be set to the maximum value of 1000 to allow for plastic hinges to develop in the pile.

Recommended:	100
Lower Limit:	10
Upper Limit::	1000

Solution Convergence

Tolerance on Solution Convergence is used by LPILE Plus to determine when the iterative solution has achieved an acceptably accurate solution. By default, this values is 0.00001 inches or meters but the solution tolerance can be changed by the user as desired.

Excessive Deflection Value

The Excessive Deflection Value is used to end analyses in which iterative solution is diverging without limit. The user should enter a value of deflection for the pile head that is grossly excessive to stop the analysis. The recommended value is 10 times the pile diameter.

Text Editor

The option of specifying a text editor program has been replaced in LPile Plus Student Edition with an internal text file viewing dialog box. This dialog can be used to view the contents of the input, output, and runtime message files.

In the professional version of LPile Plus, the user may enter the name of any program to view the output file from LPILE Plus. By default, the Notepad program included with Microsoft Windows® is the default selection. The user may find the program by pressing the Browse button on this dialog box.

The maximum size of a file that can be edited with Notepad is limited to files of 64 kilobytes. For files larger than 64 kilobytes, the user may wish to specify Microsoft Wordpad, which is usually located in the Windows Accessories folder.

Computation Menu

The commands used to run the analytical computations are found under the Computation pull-down menu. After the computations are performed successfully, this menu also provides commands to view the plain-text input data, notes produced during computations, and the computed output. This pull-down menu also provides an option for the graphical observation of the modeled pile and soil layers.

Run Analysis

An input file, after preparation or modification, will be automatically saved to disk when selecting the Computation-Run Analysis command and execute the computation portion of program LPile Plus Student Edition.

When saving data to disk, LPile Plus Student Edition will automatically add an extension of the type *.lpd to the name of the input file.

At the beginning of the run, the computation engine will read the saved input data progressively, showing the line number which is being read. If an input-data format is incorrect during reading, the analytical module will stop immediately and in many cases it saves an error message and a status report in a file with the extension *.lpr. This file may be accessed by selecting Edit Processor Run Notes in the Computation menu.

The computation engine automatically creates an output file with the same name as the input and with the extension **.lpo*. Once a successful run is produced, the user may proceed to the next items for observation of results.

View Input Text File

This menu option is used to edit the input-data file in plain-text mode using the text editor specified under the Options menu. This command becomes enabled after new data files have been saved to disk or when opening existing data files.

This command is helpful for experienced users who may just want to change one or two parameters quickly using the text editor, or for those users wishing to observe the prepared input data in text mode. However, it is recommended that users use the Windows features of the LPile Plus Student Edition to edit data files in order to eliminate easily preventable errors.

View Processor-Run Notes

This menu option is used to view an intermediate text file that is automatically produced during each analytical run. This file only includes notes produced during the processing of the input data. This menu option becomes active after new-data files have been saved to disk and analyzed or when opening previously-analyzed data files.

This menu automatically invokes the word processor or text editor that was previously specified under the Option menu. The default setting is to use the utility program named *notepad.exe* provided by Microsoft Windows ®.

Files containing processor-run notes are automatically saved to disk with the same file name as the input-data file but with the extension *.lpr.

Examination of the run notes is helpful when debugging a data file that did not result in a successful analysis.

View Output Text File

This menu option is used to view the output-text file that is automatically produced during each analytical run. This command becomes active after new data files have been saved to disk and successfully executed, or when opening previously-executed

data files.

The menu automatically invokes the word processor or text editor that was previously specified in Control Options... under the Control menu. The default setting is to use the utility program named *notepad.exe* provided by Microsoft Windows©.

Output files are automatically saved to disk with the same file name as the input-data file but with the extension **.lpo*.

Information usually contained in the output files consists of the following items, listed in order of appearance:

1. Echo-print of input data for pile analysis. Users are strongly recommended to check the echo-print of their input data to check for mistakes.
2. If selected, data for selected *p-y* curves printed for verification.
3. Data for deflection, bending moment, shear, soil resistance, and related information, as a function of depth for the modeled pile (however, this will only be printed at the level of the pile head if the user placed a check mark on Only Print Summary Table under the Options menu).
4. Output verification of finite-difference approximations, providing data about the maximum moment and lateral force imbalances observed during execution (maximum imbalances should usually consist of small numbers).
5. Output summary table, containing information about the results and number of iterations performed until convergence was reached
6. If requested, summary table of pile-head deflection versus pile length.
7. Finally, the output file will contain the summary table of data generated for pile-head foundation stiffness matrix components, if requested by the user.

Data Menu

The Data Menu allows access to the dialog boxes in which the user enters the data that goes into an LPile Plus Student Edition input data file.

Title

The title is used to identify the case being analyzed by LPile Plus Student Edition. The title may be up to 72 characters long.

Pile Properties

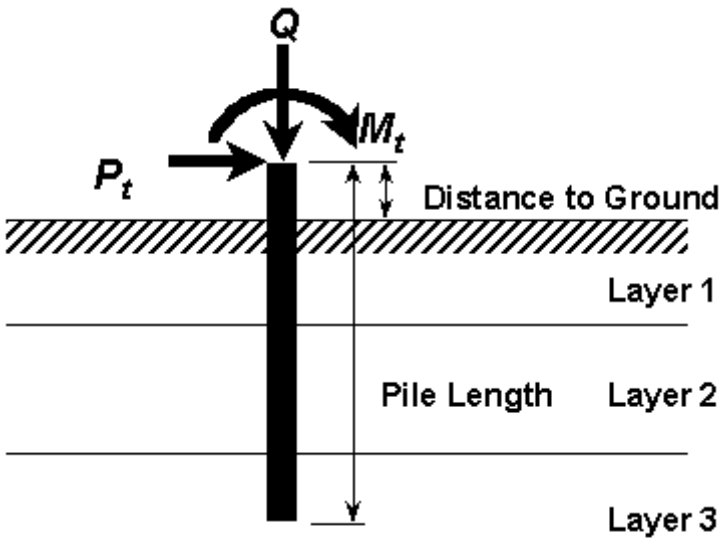
In the professional version of LPile Plus, the data for pile properties are entered in two linked dialog boxes. The first dialog box is the same for all analysis types.

The data entered in the dialog box for Pile Properties are:

Increments: The number of increments into which the pile is divided. The default value is 100 increments and the maximum allowable value is 300 increments. Use of values less than 50 increments are not recommended. The user will typically select a number of increments equal to some sub-multiple of the overall pile length, so that the nodes are spaced along the pile at conveniently round values.

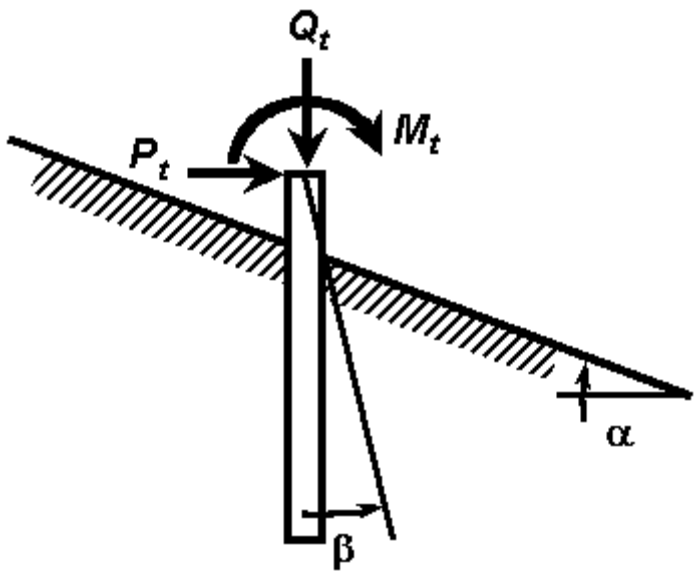
Distance to Ground: The depth below the top of the pile to the ground surface. This value should be positive in sign if the pile head is above the ground surface and negative in sign if the pile head is below the ground surface.

Pile Length: The overall length of the pile.



Note: Origin of depth coordinates for pile sections and soil layers is located at the pile head

Ground Slope: The angle, in degrees, between the ground surface and the horizontal as shown in the figure below. The slope value is positive if the pile-head moves in the downhill direction and negative if the pile moves in the uphill direction. Use of the slope value is discussed in detail in the "Line-by-line Guide to Input" in the "Guide to Preparation of Input Files" in the LPile Plus Student Edition Manual.



The following four parameters are required by LPile Plus Student Edition:

Diameter: The diameter of the pile being analyzed.

Cross-sectional Area: The cross-sectional area of the pile being analyzed.

Moment of Inertia: The moment of inertia of the pile section.

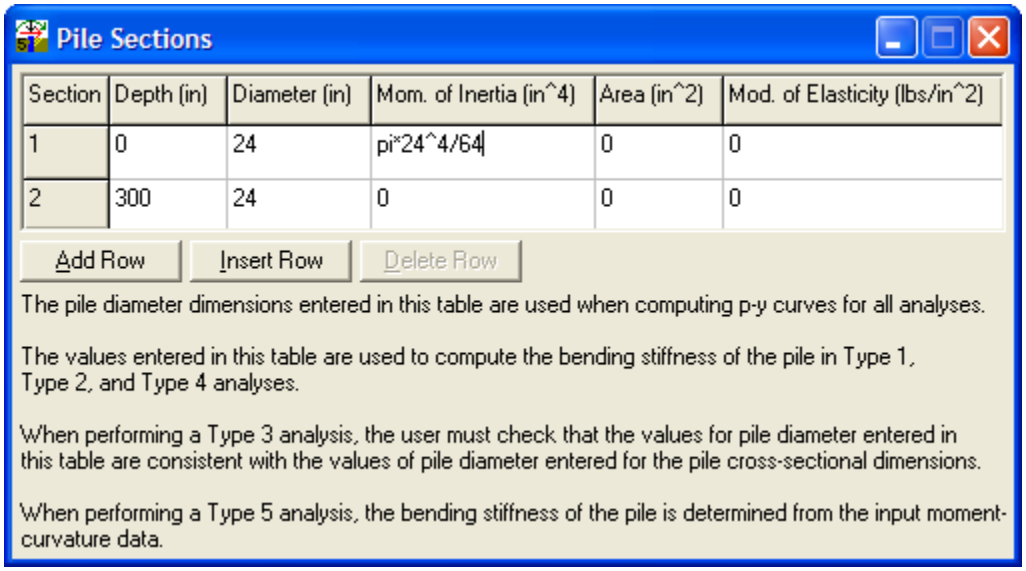
Modulus of Elasticity: The Young's modulus of the pile material.

Pile Section Properties

The Pile Section Properties table is not supported in LPile Plus Student Edition. The following information is provided to explain the use of the full professional version.

The second dialog box is activated by pressing the Edit Pile Sections button.

Note that the Pile Properties dialog box cannot be closed if important properties are missing from the Pile Sections table. The user will be informed as to what properties are missing.



The data entered in the dialog box for pile section properties are:

Top Depth: The X-coordinate (depth below top of pile) of the top or the bottom of the section for which the pile properties are specified.

Diameter: The diameter of the pile at the Top Depth location.

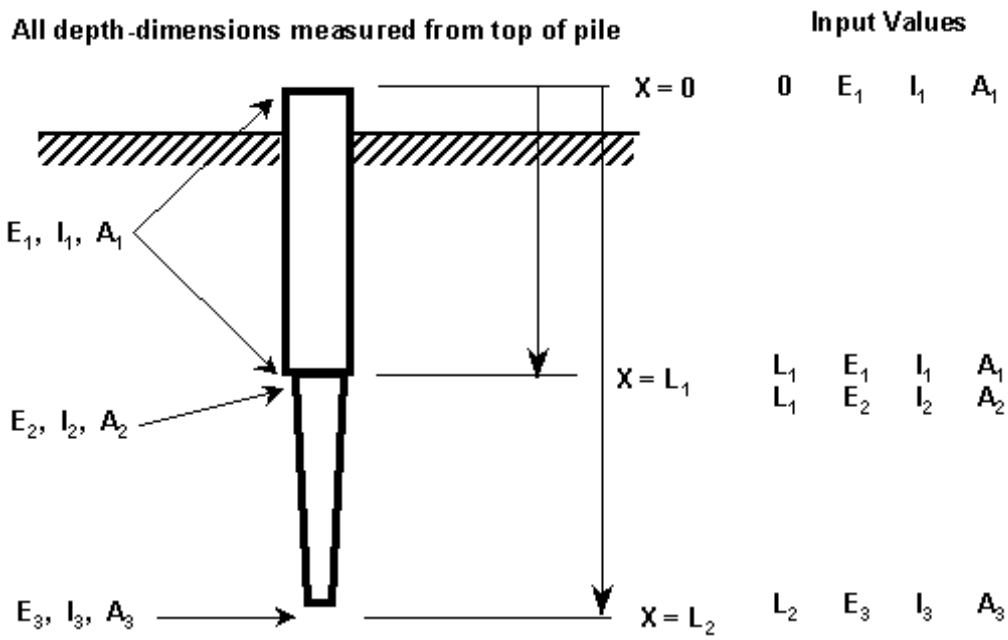
Moment of Inertia: The moment of inertia for that pile cross-section.

Area: The cross-sectional area of the pile.

Modulus of Elasticity: The modulus of elasticity.

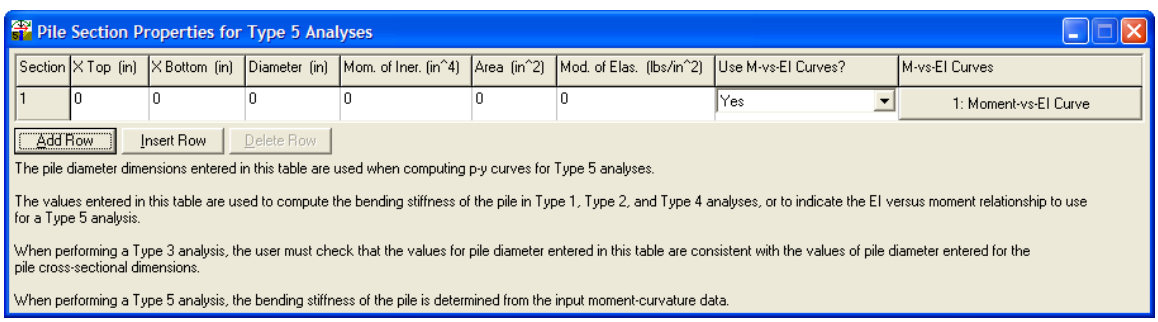
The user can enter a minimum of 2 and a maximum of 10 lines in this dialog box.

Enter values at the top and bottom of each section along the pile. The program linearly interpolates at all depths between input depths, except where step changes in properties occur.



The values entered for pile area and moment of inertia are used in computation of combined axial stress and bending. These values must be entered in all analyses, including Type 3 and Type 5 analyses that utilize nonlinear M vs. EI relationships.

For Analysis Type 5, then the dialog box is



In this dialog box, the entries for top elevation, bottom elevation, diameter, moment of inertia, area, and modulus of elasticity are the same as those entered in the basis pile sections dialog box. The link button to edit M vs. EI curves is activated when “yes is selected in the Use M-vs-EI Curves? drop down box.

Loading Type

The values entered in this dialog box are:

Type of Loading

Static Loading or Cyclic Loading

Click on the radio button appropriate to the type of loading to be analyzed. If the *p-y* curves are calculated internally and the loading is specified as being cyclic then cyclic *p-y* criteria will be used.

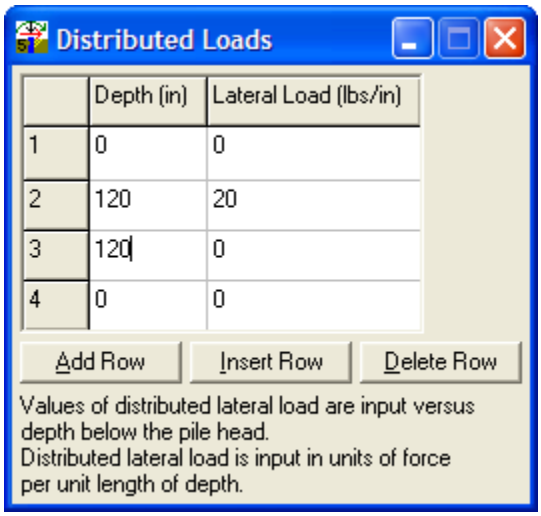
Number of Cycles

This entry field is activated only if the cyclic loading radio button is selected. The minimum valid value is 1 and the maximum valid value is 5,000 cycles of loading.

Distributed Loads: This feature is not supported in LPile Plus Student Edition. Check the box for distributed loads if distributed lateral loads are to be applied to the pile. The dialog box for entering the distributed lateral loading is activated by pressing the Edit Distributed Loads button. The distribution of lateral load is entered using a minimum of 2 and a maximum of 10 points. The depth values are measured from the pile head and the lateral load values are entered in units of force per length (i.e. load intensity).

Distributed Loads

The Distributed Loads table is not supported in LPile Plus Student Edition. The following information is provided to explain the use of the full professional version.



In this dialog box, the user may enter from 2 up to 10 lines of distributed loading data.

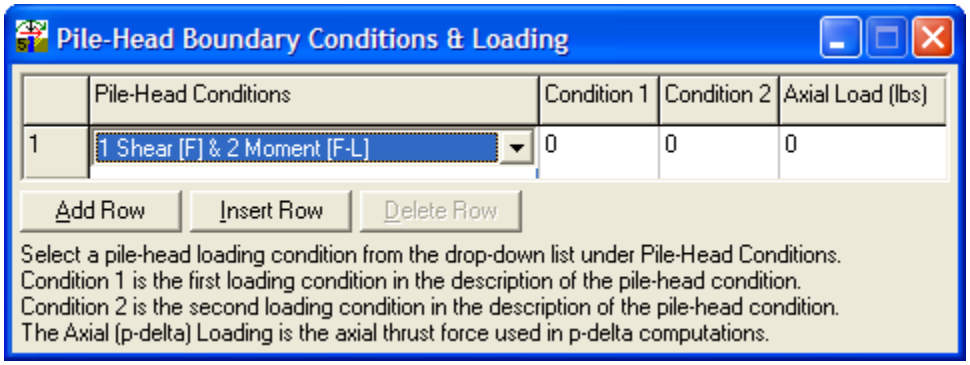
The two entry fields per line are:

Depth: The depth (L) below the top of the pile where the distributed load is specified. Enter Depths in ascending order.

Loading: The distributed lateral load (Force/Length) on the pile. A value of 0 is assumed if this field is left blank.

Pile-Head Boundary Conditions and Loading

In this dialog box, the user may enter up to 10 lines of loading conditions (also known as boundary control conditions). The four fields per line are:



Pile-Head Conditions:

A drop-down box that controls boundary conditions at the pile head. There are four options, which select the following paired specified values:

1. Lateral shear force and moment at the pile head (moment = 0 for a free-head pile).

2. Lateral shear force and slope at the pile head, in radians (slope = 0 for a fixed head pile).

3. Lateral shear force and rotational restraint at the pile head.

4. Lateral displacement and moment at the pile head.

5. Lateral displacement and slope at the pile head, in radians (slope = 0 for a fixed head pile).

Value for Condition 1: The value entered in this column can have one of two different values depending on the pile-head boundary conditions specified:

For pile-head conditions 1, 2, and 3: Enter the lateral load (F) at the top of the pile.

For pile-head conditions 4and 5: Enter the horizontal displacement (L) at the top of the pile.

Value for Condition 2: The value entered in this column can have one of three different values depending on the pile-head boundary conditions specified:

For pile-head conditions 1 and 4: Enter the moment (F-L) at the top of the pile.

For pile-head conditions 2 and 5: Enter the slope (dimensionless) at the top of the pile.

For pile-head condition 3: Enter the rotational stiffness (F-L / radian).

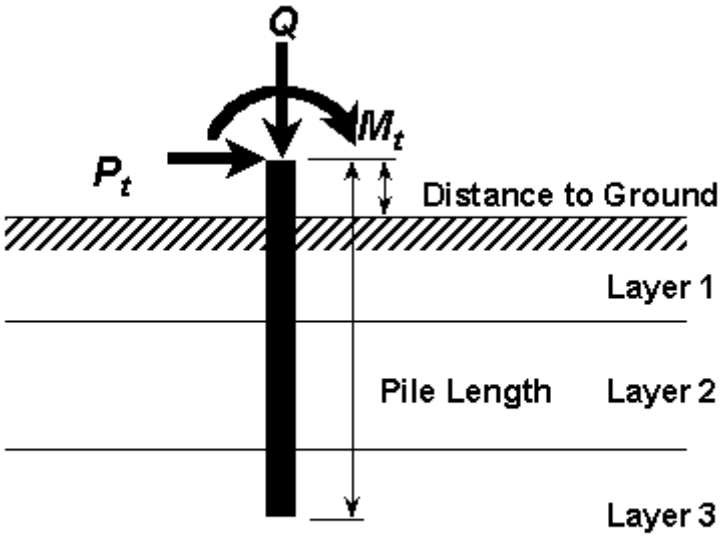
Axial Load: The axial load (F) at the top of the pile, in terms of the units of force, positive values act downwards in compression.

Sign Convention for Loads:

Vertical forces are assumed to be positive acting downwards.

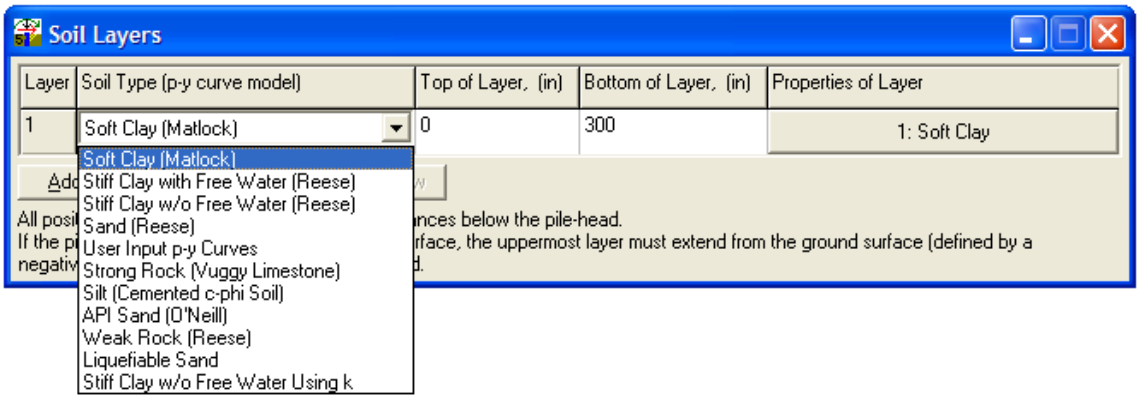
Horizontal forces are assumed to be positive acting to the right.

Moments acting on the foundation are assumed to be positive acting clockwise.



Note: Origin of depth coordinates for pile sections and soil layers is located at the pile head

Soil Layering and *p-y* Curves



This dialog box allows entry and editing of the data that describe the soil profile. These variables are:

Soil Type: The soil type for each layer can be selected using the drop-down box for soil type.

Layer Top: The elevation coordinate of the top of the layer (measured from the pile head).

Layer Bottom: The elevation coordinate of the bottom of the layer (measured from the pile head).

Values for the upper elevation coordinate for the surface layer can be negative in sign if the pile head is below the ground surface.

The Student Edition is limited to single layer profiles only.

Required Parameters for *p-y* Curves

The text displayed on the table button will change depending on the selected soil type. The table buttons activate data entry dialog boxes that are unique for each selected soil type. These dialog boxes provide the means to enter effective unit weight, shear strength parameters, and any other required soil property parameters. The following parameters are required by more than one *p-y* type.

Effective Unit Weight

Values of effective unit weight are required for computation of vertical effective stress, which is used in computation of *p-y* curves. Values are input at the top and bottom elevations of a layer and values are determined by linear interpolations between these two elevations.

Values at elevations between the top and bottom elevations will be determined by linear interpolation.

Cohesive Strength

Shear Strength (F/L2): The undrained shear strength of cohesive soil at the specified depth.

Values at elevations between the top and bottom elevations will be determined by linear interpolation.

Angle of Friction

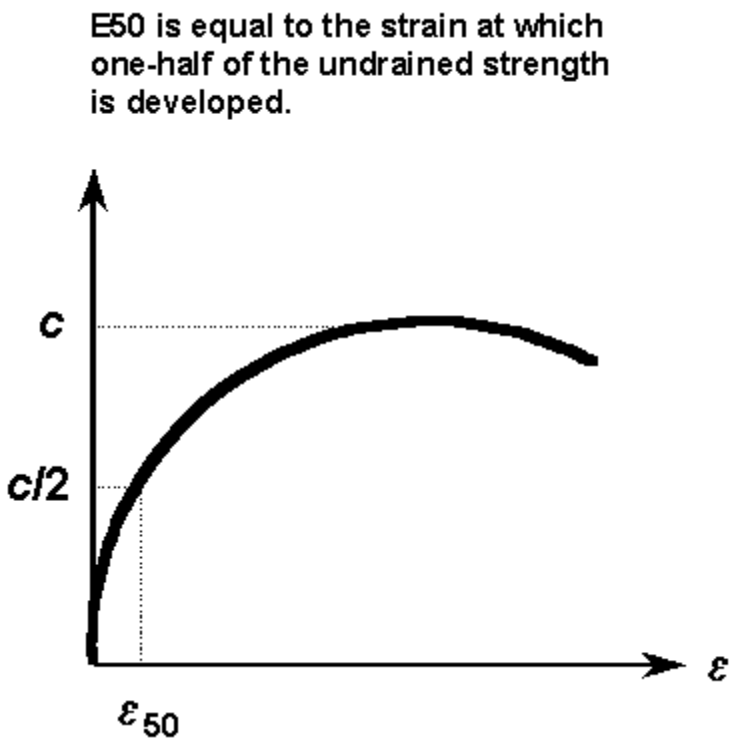
Angle of Internal Friction (degrees): The angle of internal friction, ϕ, in degrees, for the specified depth.

Values at elevations between the top and bottom elevations will be determined by linear interpolation.

Soil Strain Parameter E50 for Clay Soils

The E50 parameter is obtained either from laboratory testing or by correlation with undrained shear strength. The input value E50 represents the axial strain at which 50 percent of the undrained strength is developed in a laboratory compression test as shown in the figure below.

This value is also referred to as ε_{50}.



Internal default values of E50: LPile Plus will use default values of E50 if the input value is left equal to zero.

The default values are the values shown below:

Consistency	Value of *c*	E50Value
Soft	1.74 - 3.47 psi 250 - 500 psf 12 - 24 kPa	0.02
Medium	3.47 - 6.94 psi 500 - 1,000 psf 24 - 48 kPa	0.01
Stiff	6.94 - 13.9 psi 1,000 - 2,000 psf 48 - 96 kPa	0.007
Very Stiff	13.9 - 27.8 psi 2,000 - 4,000 psf 96 - 192 kPa	0.005
Hard	27.8 - 55.6 psi 4,000 - 8,000 psf 192 - 383 kPa	0.004

Soil Modulus Parameter *k* for Clays

k-value: The soil parameter *k* (F/L^3) is used in the equation $E_s = k\ x$ where E_s is the secant modulus on a *p-y* curve and *x* is the depth below the ground surface. The value of *k* describes the variation (increase) in E_s with depth. If left equal to zero LPile Plus will

use the following values by default.

Consistency	Avg. Undrained Shear Strength (*c*)	Static Loading	Cyclic Loading
Soft	1.74 - 3.47 psi 50 - 500 psf 12 - 24 KPa	30 pci 8,140 KPa/m	- -
Medium	3.47 - 6.94 psi 500 - 1,000 psf 24 - 48 kPa	100 pci 27,150 kPa/m	- -
Stiff	6.94 - 13.9 psi 1,000 - 2,000 psf 48 - 96 kPa	500 pci 136,000 kPa/m	200 pci 54,300 kPa/m
Very Stiff	13.9 - 27.8 psi 2,000 - 4,000 psf 96 - 192 kPa	1,000 pci 271,000 kPa/m	400 pci 108,500 kPa/m
Hard	27.8 - 55.6 psi 4,000 - 8,000 psf 192 - 383 kPa	2,000 pci 543,000 kPa/m	800 pci 217,000 kPa/m

Note that the *k*-value is not the same as the modulus of subgrade reaction used to describe elastic settlements of shallow foundations. The *k*-value is determined either from analysis of full-scale load test of piles subjected to lateral loading or from the correlations provided by the entries above.

The *k*-value is used to determine the initial slope for the *p-y* curve for soil types 2 and 4. It is used to define initial soil moduli of the *p-y* curve for the first iteration of the solution for the other soil types.

Soil Modulus Parameter *k* for Sands

If left equal to zero, LPile Plus will use the following values by default.

Relative Density	*Loose*	*Medium*	*Dense*
Sand Below Water Table	20 lb/in^3 5,430 kPa/m	60 lb/in^3 16,300 kPa/m	125 lb/in^3 33,900 kPa/m
Sand Above Water Table	25 lb/in^3 6,790 kPa/m	90 lb/in^3 24,430 kPa/m	225 lb/in^3 61,000 kPa/m

Required Properties for Soft Clay

See the prior discussion for effective unit weight, undrained cohesion, and E50.

Required Properties for Stiff Clay with Free Water

The criteria for stiff clay with free water is intended for use with stiff clays that may be

subject to soil erosion around if any gap forming around a pile is filled with water.

See the prior discussion for effective unit weight, undrained cohesion, E50, and *k*.

Required Properties for Stiff Clay without Free Water

The criteria for stiff clay without free water is intended for use with stiff clays that are not subject to soil erosion around if any gap forming around a pile is filled with water.

See the prior discussion for effective unit weight, undrained cohesion, and E50.

Required Properties for Stiff Clay without Free Water Using Initial *k*

The criteria for stiff clay without free water is intended for use with stiff clays that are not subject to soil erosion around if any gap forming around a pile is filled with water.

This criteria is tended for use only in areas where load test results have demonstrated that the standard method for stiff clay without free water yields a *p-y* curves that is too stiff at small lateral deflections. The value of *p-y* modulus to enter should be determined from lateral load tests.

The required properties are: effective unit weight and undrained cohesive strength. The optional properties are *p-y* modulus and the strain factor E50 (ε_{50}).

Required Properties for Sand and API Sand

See the prior discussion for effective unit weight, undrained cohesion, friction angle, and *k*.

Required Properties for Liquefied Sand

The only required input for the *p-y* curve in liquefied sand is effective unit weight.

Note that it is not possible to perform an analysis with liquefied sand in LPile Plus Student Edition.

Required Properties for Cemented Silt

See the prior discussion for effective unit weight, undrained cohesion, friction angle, *k*, and E50

Required Properties for Vuggy Limestone

See the prior discussion of effective unit weight. The uniaxial compressive strength of the rock core differs from the undrained cohesive strength by the fact that it represents the diameter of the Mohr's circle at failure, rather than the radius of the Mohr's circle.

Required Properties for Weak Rock

See the prior discussion for effective unit weight and the following sections for other parameters.

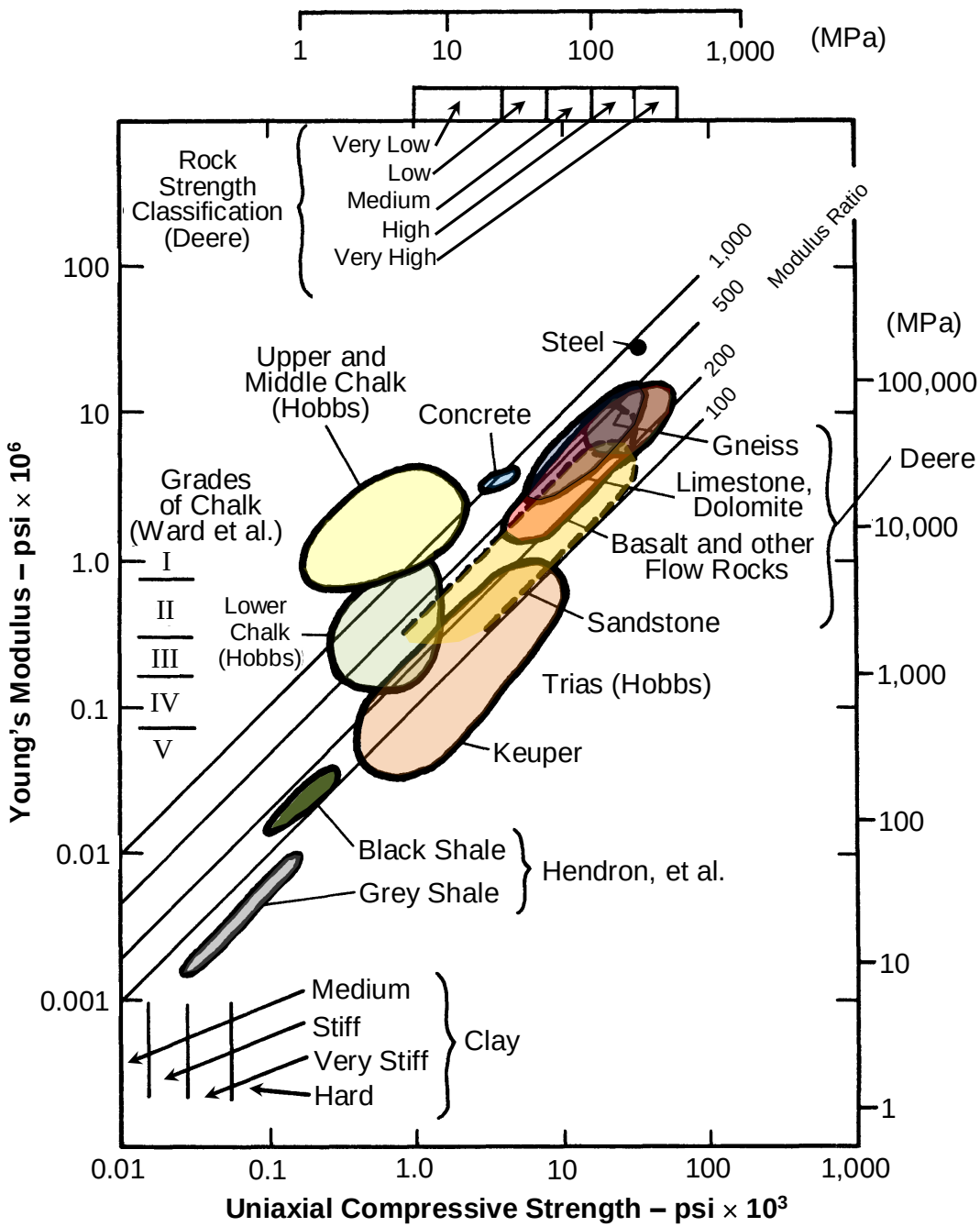
Elastic Modulus for Weak Rock

The rock mass modulus for weak rock should be entered for this value.

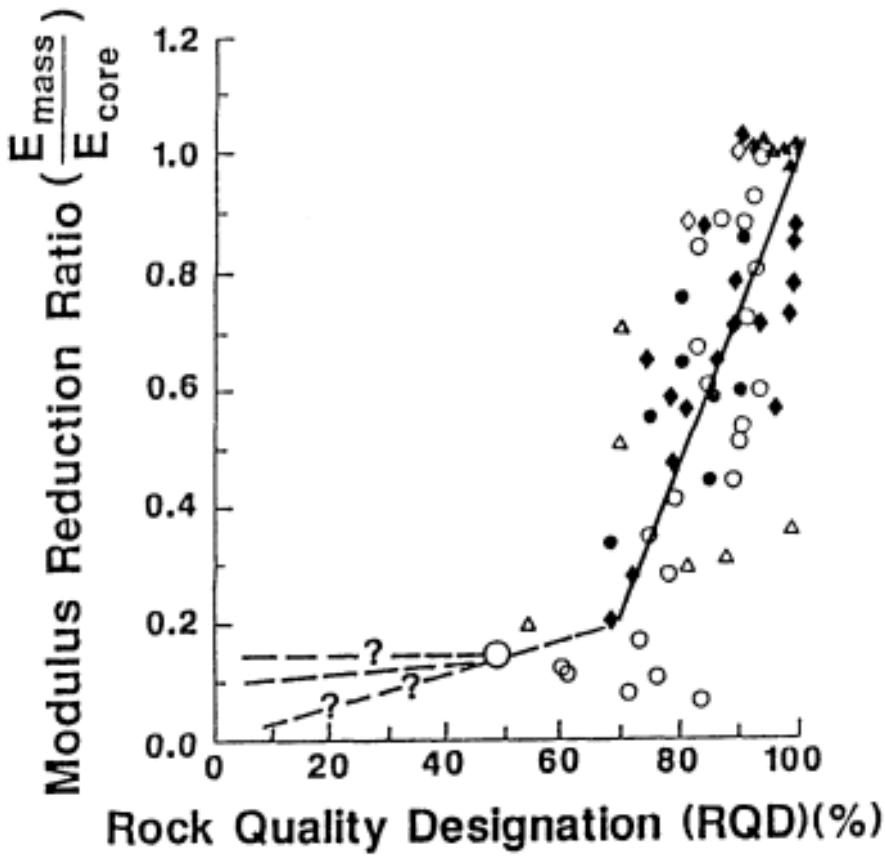
The units of input will be Force per Length squared.

This rock mass modulus may be measured in the field using the initial slope of a pressuremeter test.

Alternatively, the rock mass modulus may be obtained from the product of the modulus reduction ratio and Young's modulus of intact rock specimens tested in uniaxial compression. The chart developed by Horvath and Kenney, shown below, may be used to obtain the intact rock modulus by correlation with uniaxial compressive strength and rock type.



The chart developed by Bienawski, shown below, may be used to obtain the modulus reduction ratio. Note that the correlation for modulus reduction ratio is uncertain for RQD values of less than 50 percent. If possible, the user of the program should consult with those who have examined the weak rock specimens in the field to assess whether the rock mass ratio should be estimated at the upper or lower portions of the range indicated by the figure below



Uniaxial Compressive Strength

The uniaxial compressive strength of the rock core differs from the undrained cohesive strength by the fact that it represents the diameter of the Mohr's circle at failure, rather than the radius of the Mohr's circle.

Rock Quality Designation

The fracturing of the rock is described using the Rock Quality Designation (RQD). The value of RQD is input as a percentage. RQD is computed by adding the lengths of all intact core sections greater than 4 inches (100 mm) in length and dividing by the total core run. An input value of 100% indicates solid rock.

Modulus Parameter k_{rm} for Weak Rock

The modulus k_{rm} for weak rock defines the initial modulus of the *p-y* curve for weak rock.

This parameter is based on the axial strain developed in a uniaxial compression test on weak rock. This parameter is equal to the axial strain developed at a stress level of 50

percent of the uniaxial compressive strength. This parameter is similar to the Soil Strain Parameter E50 used for clay soils.

This parameter is determined in the same manner as Soil Strain Parameter E50 for clays. If laboratory data are not available, values ranging from 0.00005 to 0.0005 are suggested.

Typical values of k_{rm} range from 0.00005 to 0.0005.

Required Properties for User-Input *p-y* Curves

Enter the effective unit weight at the top and bottom elevations for the soil defined by user input *p-y* curves as shown in the upper dialog box. The top and bottom of layer depths values must be entered in ascending order (i.e. from the pile top to pile bottom).

The pile head is the origin of the depth coordinates. Thus, a depth of zero is located at the pile head.

The link buttons activate the dialog boxes for input of the user input *p-y* curves. Each individual *p-y* curve may have up to 15 data points. The user must define two *p-y* curves at the top and bottom of the specified layer. Additional soil layers may be specified to cover a suitable range of depths so that the lateral load-transfer data is adequately defined over the length of the pile.

LPile Plus Student Edition will linearly interpolate for *p*-values over the depth between input *p-y* curves.

The user should enter identical *p-y* curves at the top and bottom of a soil layer should the user desire the *p-y* curves be uniform over that layer.

The values comprising each pair of data are:

Deflection: The lateral pile deflection of a point on a *p-y* curve.

Soil Resistance: The soil resistance in units of force/length.

Enter these data pairs so that the deflection values are in ascending order.

Modification Factors for *p-y* Curves

Modification factors for *p-y* curves are used to model the lateral load-transfer behavior of individual piles in a pile group. Three input values are required when use of *p-y* multipliers has been selected under the Options Menu. The required values are the depth at which the *p*-multiplier and *y*-multiplier are defined and values of the *p*- and *y*-multipliers. The *p*-multiplier is used to reduce the ultimate soil resistance. The *y*-multiplier is used to increase pile deflection due to soil elastic interaction.

In this dialog box, three values are entered by the user for each *p-y* modification factor. They are:

Depth of Multiplier: The X-coordinate: the depth below the top of the pile.

p-Multiplier: Reduces the ultimate soil resistance.

y-Multiplier: Increases the pile deflection due to soil elastic interaction.

The user can enter a maximum of 30 lines in this dialog box.

Please note that the manner in which the *p*- and *y*-multipliers are implemented in an

analysis is the following. First, compute the value of *p* for a given value of *y* using the specified *p-y* criteria and compute the *p-y* modulus by dividing *p* by *y*. Secondly, values of the *p*- and *y*-multipliers are obtained by interpolation for the depth of interest. Lastly, the *p-y* modulus of the first step is multiplied by the *p*-multiplier and divided by the *y*-multiplier. Thus, if the *p*- and *y*-multipliers are equal in value, then there is no effect on the computed results.

Shearing Resistance Curve at Pile Tip

In this dialog box, the user inputs data to describe the shear force versus displacement curve at the pile tip. The two input variables are:

Displacement: The displacement at the pile tip. It is in terms of inches, meters or any consistent unit of measurement, as selected in the *Options: Units* dialog.

Shear Resistance: The amount of shear resistance experienced for the given displacement, in pounds, kilonewtons or other consistent unit of force.

The user must enter at least 2 and no more than 20 lines in this dialog box.

If the pile is long and lateral movement of the pile tip is near zero, the pile tip resistance curve will have little effect on the results of computation. The usual use for the feature is in analyzing the results of lateral load tests on short piles in which substantial movement of the pile tip has occurred. In this situation, the pile tip resistance curve is varied until pile-head displacement and rotation match the values measured in the lateral load test. This is repeated for all loads applied in the lateral load test until the full range of data for the tip shear resistance curve has been developed.

The pile tip resistance curve is usually proportional to the cross-sectional area of the pile tip. Thus, any pile tip resistance curve must be adjusted for any change in cross-sectional area when analyzing piles other than the test pile.

No method for computing the tip shear resistance curve is widely accepted. Usually, the values for this curve are determined from lateral load tests on short piles or drilled shafts.

Cross-Section Analysis

The full professional version of LPile Plus has the capability to evaluate the ultimate moment capacity and nonlinear bending stiffness relationships for deep foundations made from normally-reinforced concrete, pipe sections, and prestressed concrete. These features can determine how the effective bending stiffness will vary as the concrete cracks in tension and the reinforcing steel yields.

Use of the features to evaluate ultimate moment capacity and nonlinear bending stiffness is essential when analyzing the behavior of drilled shafts under lateral loading.

Suggested Method for Using Nonlinear *EI*

The features supporting the use of nonlinear moment-curvature relationships are not

supported in LPile Plus Student Edition.

The following is a brief description of the method suggested by Ensoft for performing analyses that include the effects of nonlinear bending.

Proper use of the features for cross-sectional analysis require that the designing engineer select an appropriate factor of safety or set of load and resistance factors to use when evaluating whether the computed level of bending moment is acceptable for design. This decision must be made by the designing engineer before the design analysis can proceed. More information is available for Structural Resistance Factors and Design Load Combinations

The first step is to do a cross-section analysis for the combinations of pile dimensions and reinforcing steel details that may be used. These analyses are performed by selecting the second option in the Options - Analysis Type dialog box. From the graphs of *EI* versus moment and bending moment versus curvature, the user can select trial values of pile modulus and allowable bending moment for use in analyses performed in the second step of the design. The modulus of elasticity for the pile is computed by determining the *EI* value for the level of bending moment then dividing the *EI* value by the moment of inertia of the cross-section.

For the second step, it is usually preferred to determine the initial size of the pile using elastic bending properties before performing the analysis that includes the effects of nonlinear bending. The reason for this is that this approach allows the user to determine whether the soil is capable of supporting the loading on a pile of a given diameter and length. To do this, the use selects the first option in the Options - Analysis Type dialog box.

When performing the elastic analysis, the user should check that the pile is sufficiently long such that the computed bending moment at the pile tip is close to zero.

Before performing the third step in the design analysis, the user should compare the maximum mobilized bending moment obtained in the first step analysis to the ultimate moment capacity obtained in the second step analysis, while applying the appropriate factor of safety or load and resistance factors. If the ultimate moment capacity is too low for the maximum mobilized bending moment, steps 1 and 2 should be repeated using a stronger pile cross section.

The third step in the design analysis is the analysis in which the nonlinear *EI* is included. Once the approximate diameter and length of the pile has been determined, the user then selects the third option in the Options - Analysis Type dialog box to include the effects of nonlinear bending in the analysis.

When performing the analyses using the third analysis type, the user must check the magnitude of pile-head displacements and rotations and that a plastic hinge is not being developed in the pile. It is possible that the computed displacements and rotations are unacceptably large, even though the moment in the pile is under the allowable bending moment. In this event, the diameter and possibly the length of the pile must be increased in order to reduce the computed displacements and rotations.

A plastic hinge may develop if the pile is overloaded. Usually, this is determined when the maximum bending moment exceeds the allowable bending moment, and/or when the limit of iterations allowed is reached. A “push-over analysis can be performed in order to identify the level of loading that may result in a plastic hinge is developed. A push-over analysis is performed by using a set of prescribed pile-head displacement

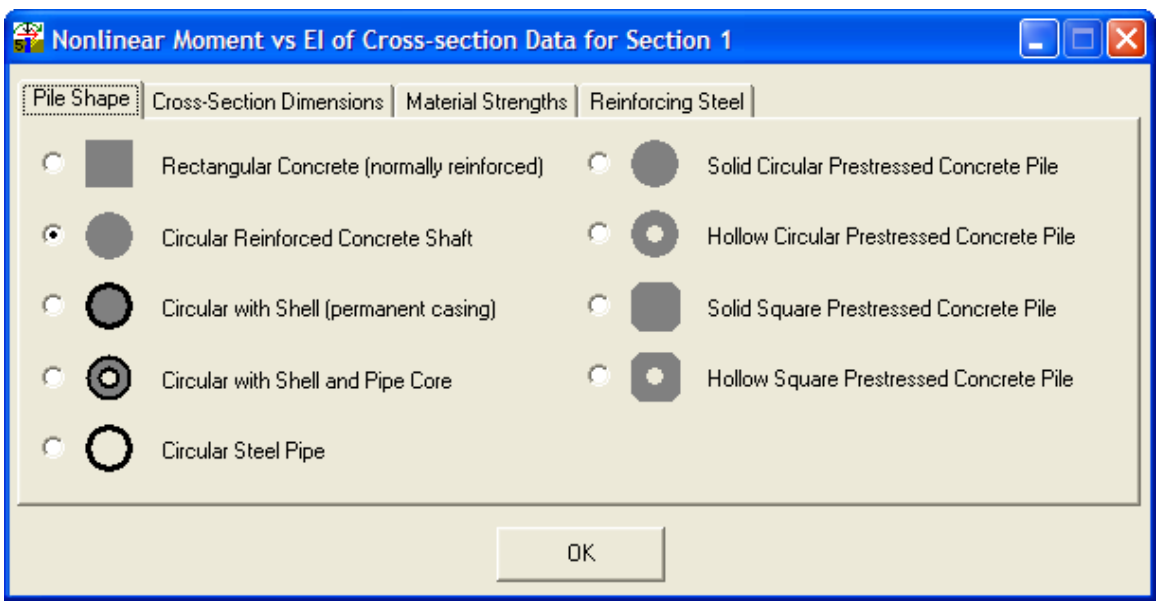
loading conditions. The pile-head displacements should cover a range of displacements from those for which the pile does not fail to several for which a plastic-hinge does develop. The magnitude of loading can be easily identified by examining the graph of pile-head displacement versus pile-head load and the location of the plastic hinge can be identified by examining the graph of lateral pile displacement versus depth. The location of the plastic hinge is the point below which the pile response curves overlap and above which the curves diverge.

If the computed lateral displacement and bending moment is near zero over a substantial fraction of the pile above the pile tip, it may be possible to shorten the length of the pile, though often the length of embedment is determined by the length required for axial loading.

If the computed lateral displacement at the pile tip is not close to zero (this is called "fence post behavior), it is usually desirable to lengthen the embedded length of the pile until computed lateral displacement is near zero at the pile tip.

An easy way to determine the optimum embedded length of pile is to use the analysis option for automatically generating pile top displacement vs. pile length.

Cross-Sectional Shapes



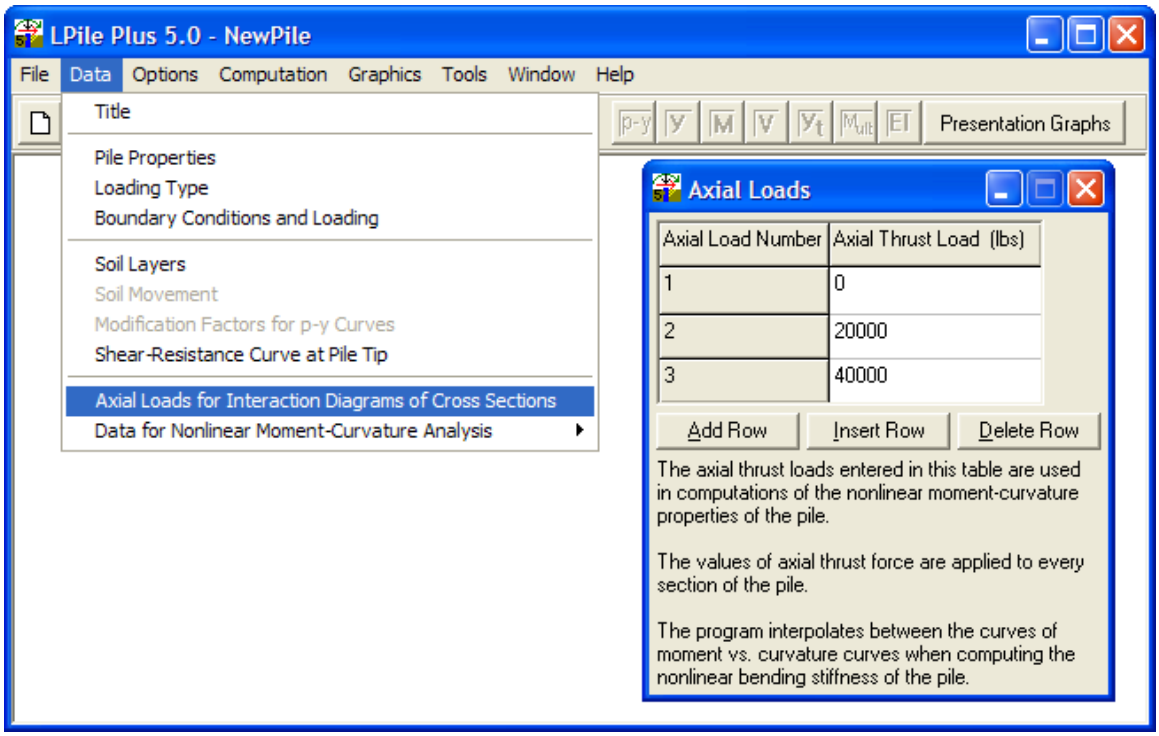
LPile Plus can perform calculations for nine different cross sections. These sections are:

1. Rectangular or square section,
2. Circular without shell or core,
3. Circular with shell but without core,
4. Circular with shell and core,
5. Circular steel pipe,
6. Solid round prestressed pile,
7. Hollow round prestressed pile,

8. Solid square prestressed pile with chamfered corners, and
9. Hollow square prestressed pile with chamfered corners.

The following sketch illustrates these nine structural shapes.

Axial Loads for Cross-Section Analysis



This dialog box is used to enter the axial loads for which non-linear moment-curvature behavior is computed. The range and magnitude of the axial loads specified differ depending on whether a lateral load analysis using non-linear bending is being performed or an interaction diagram is being computed. The total number of axial loads that can be specified is 10.

Note that the same set of axial loads is used for every section of the pile or drilled shaft when computed the nonlinear bending relationships.

If a lateral load analysis is being performed, the user should enter values of axial loads that correspond to the range of axial loads specified under Boundary Conditions and Loading. LPile Plus Student Edition will interpolate between the generated curves on the basis of axial force specified for the problem and computed bending moment.

When multiple axial loads are specified, the program can extract the maximum allowable moment capacity for each axial load and generate an unfactored interaction diagram.

If an interaction diagram is desired, the user has to specify the axial loads used to generate the diagram. The maximum axial thrust specified for an interaction diagram will depend on the structural type of the pile and the building code for which the structure is being designed.

The interaction diagram computed by the program is approximate and is not factored for magnitude of axial thrust forces because the strength reduction factors used depend on

the type of transverse reinforcement used in the reinforced concrete member. It is recommended that the user consult Sections 9.3.2.2 and 10.11 of ACI 318-95 or later to obtain the strength reduction factors for axial load with flexure if the user wishes to compute a factored interaction diagram.

Graphics

The Graphics menu is used to generate and observe plots of the results of a successful analysis. Options for the observation of results this menu are enabled only after a successful run has been made. Even after performing successful runs, some options may still be disabled since the type of output data depends on options selected for each analysis.

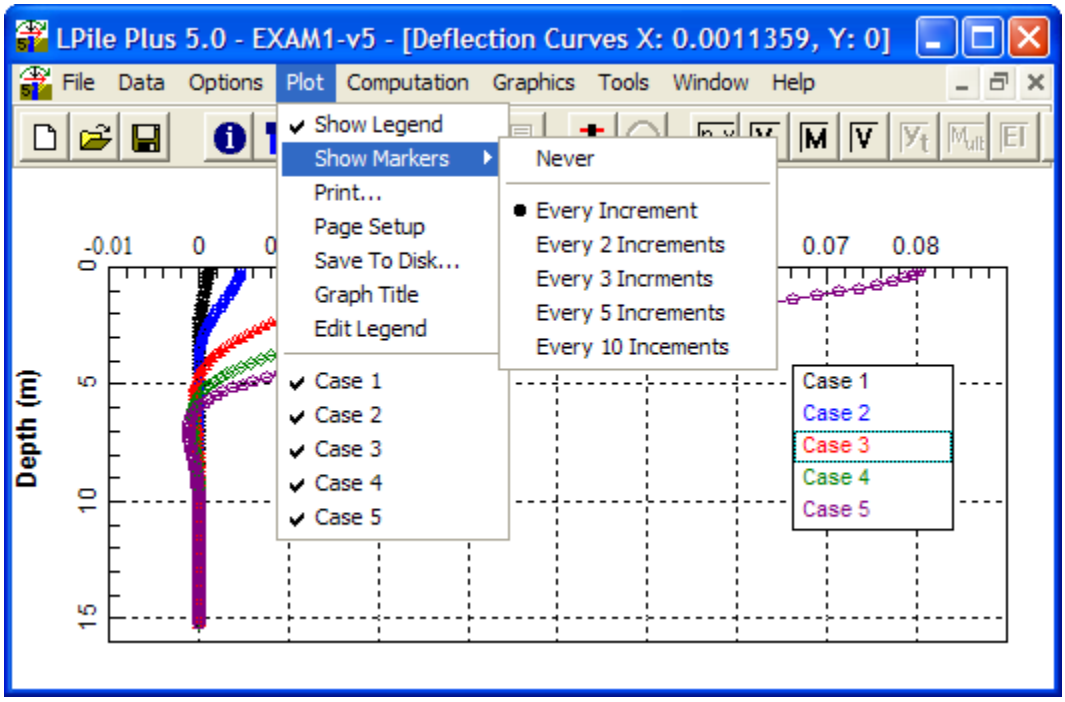
All graphs generated by LPILE Plus have a zooming feature that can be used to enlarge or reduce the size of the plotted area on the screen.

To zoom in on a portion of the graph, position the mouse at one corner of the selected area, press the left mouse key, and drag to the opposite corner.

To zoom out, press the right mouse key.

When a legend is displayed, the legend may be moved around the plot area by dragging with the left mouse button.

When multiple curves are displayed, the display of individual curves can be triggered on or off by clicking on the color-coded label in the legend for the graph.



Plot Menu – Modify, Save, and Print Graphics

This Plot pull-down menu only appears in the top-menu bar while in the graphics mode. Command options contained in this menu are shown and described below.

Show Legend

This activates or deactivates the floating legend included in the graphics mode. The legend box can be moved with click-and-drag operations of the mouse to any position in the graphics screen.

Show Markers

This activates or deactivates the markers for each point of all the curves displayed in the active screen of the graphics mode. The type of markers used in each curve is automatically selected by the program. Once activated, the markers are enabled for all visible curves of the active graphics screen.

Print Graph

This command is used to produce a hard copy of the active graphics screen. The Print sub-windows includes a drop-down menu that allows the user to select any of the installed Windows printers, allows for change of printer properties, and allows the selection of the total number of copies to be printed. The printed graphics is automatically sized to occupy the maximum width of the selected page.

Save Graph to Disk

This command is used to save the current graphics screen to disk. The format of the saved file is as *bmp* graphics (bitmap file), which is the internal-file format used in Microsoft Windows ©.

Graph Title

This command is used to activate a dialog box in which the user may enter a title to be displayed for the graph. The user may also indicate whether to locate the title at the top or bottom of the graph.

Edit Legend

The command is used to edit the labels used in the legend displayed with the graph.

Pile-Soil Geometry

This command is used to display a side-view image of the pile and soil profile.

Printing of the pile-soil geometry is possible by the print command shown in the dialog box.

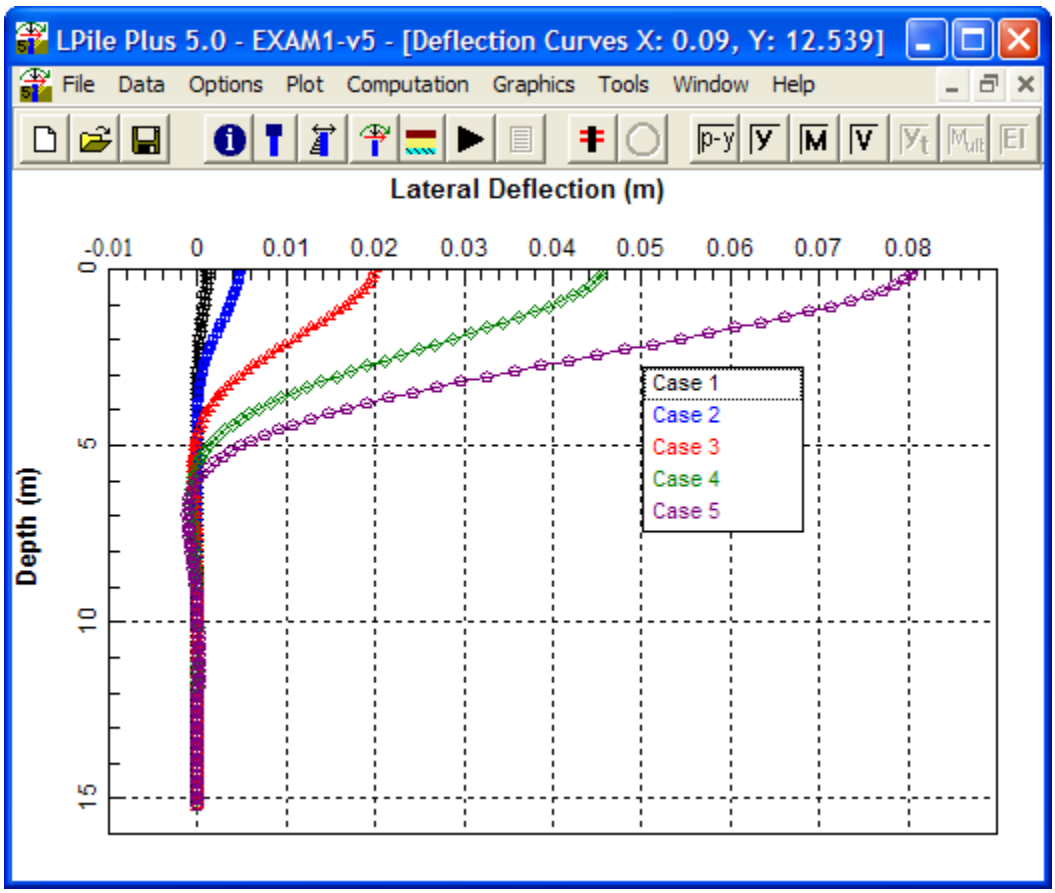
The image shown indicates the length and dimensions of the pile, the soil layering, and the combined angle for ground slope and pile batter (note that a battered pile is always displayed as a vertical pile in the image). Each soil layer is displayed using a hatch pattern that is determined by the type of *p-y* curve chosen for the soil layer.

The display of the hatch pattern can be turned off by checking the Omit Hatches check box.

p-y Curves

LPILE Plus is capable of generating *p-y* curves at user-specified depths. This graphics option only becomes available if the user asked the program to print *p-y* curves for verification purposes using the command Print *p-y* Curves under the Options menu (see section 3.3.10 of the *User's Manual* for further references). When specified, the graphics screen will show the *p-y* curves for all selected depths. When multiple curves are displayed, the display of individual curves can be triggered on or off by clicking on the color-coded label in the legend for the graph.

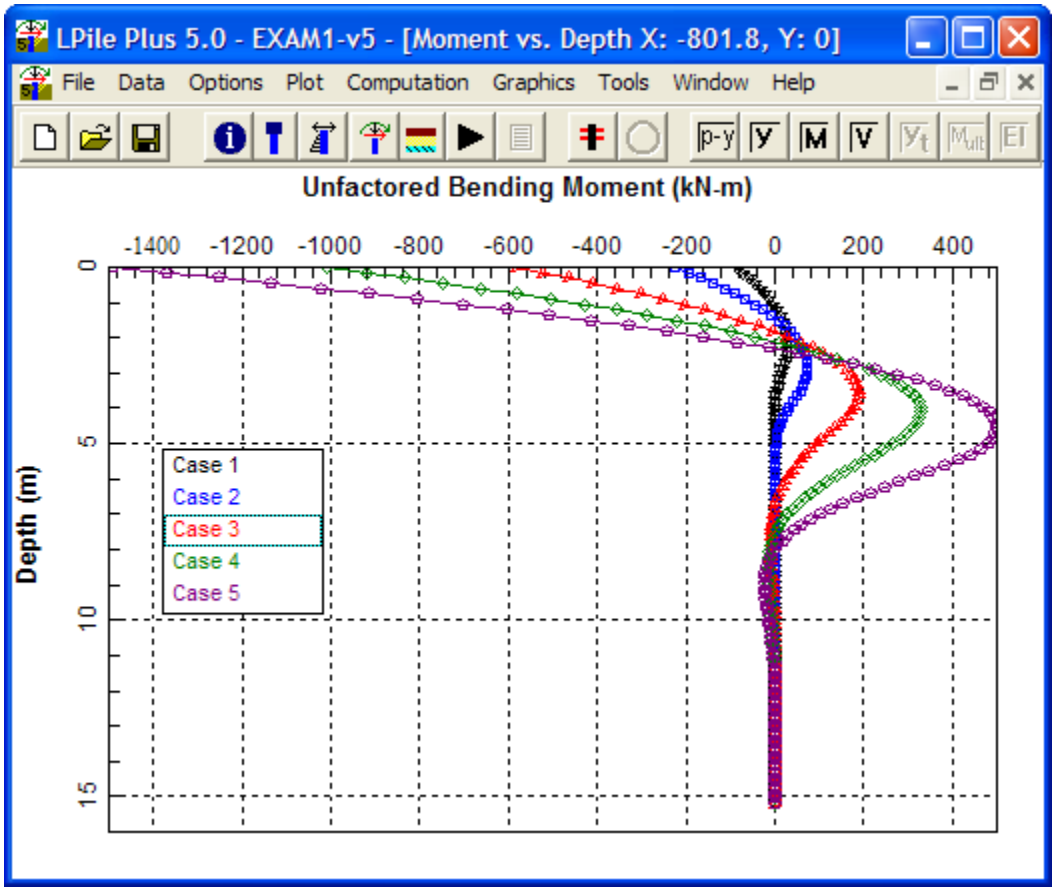
Lateral Pile Deflection vs. Depth



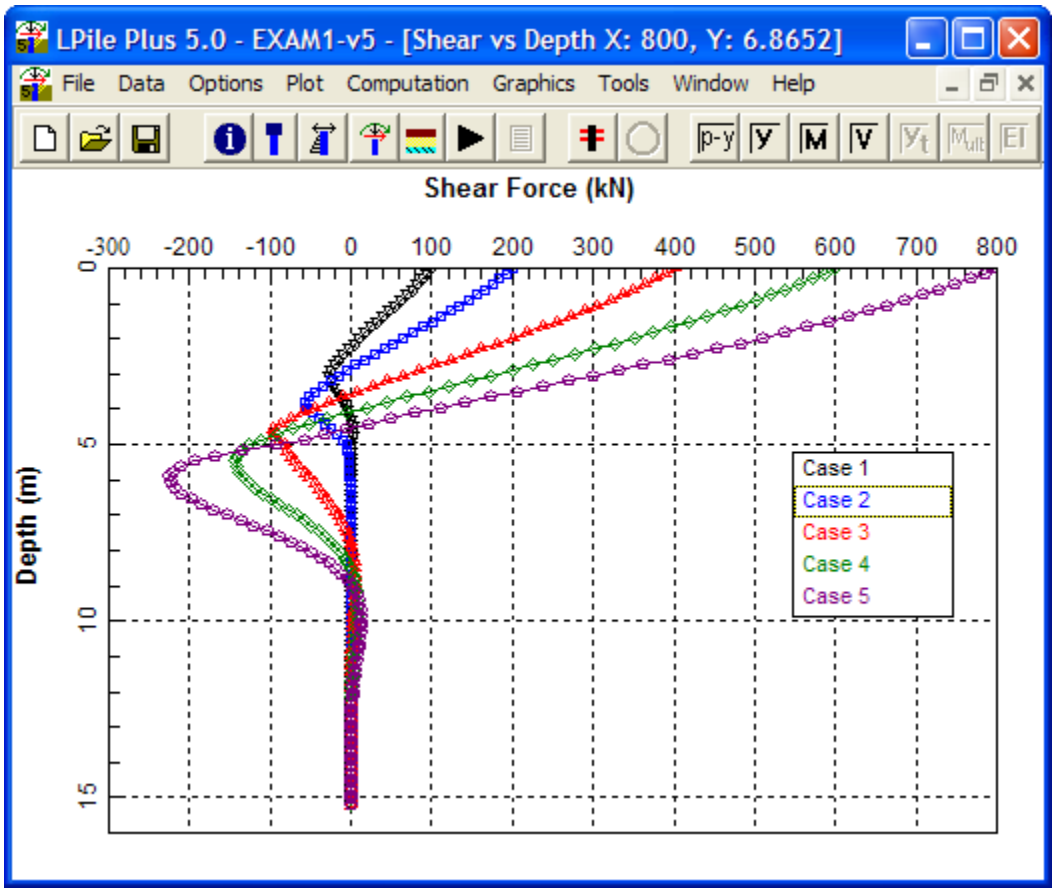
The user may select this command option to observe a graphical representation of the curve of lateral deflection versus depth for the modeled pile. This curve is automatically generated in all analytical runs of a laterally loaded pile. The number of points on the deflection curve is equal to the selected number of pile increments. Several curves may be contained in the graphics if the user selects to input several load cases. When multiple curves are displayed, the display of individual curves can be triggered on or off by clicking on the color-coded label in the legend for the graph.

Mobilized Bending Moment vs. Depth

The user may select this command option to observe a graphical representation of the curve of bending moment versus depth for the modeled pile. This curve is automatically generated in all analytical runs of a laterally loaded pile. The number of points on the moment curve is equal to the selected number of pile increments. Several curves may be contained in this graphics if the user selects to input several load cases. When multiple curves are displayed, the display of individual curves can be triggered on or off by clicking on the color-coded label in the legend for the graph.

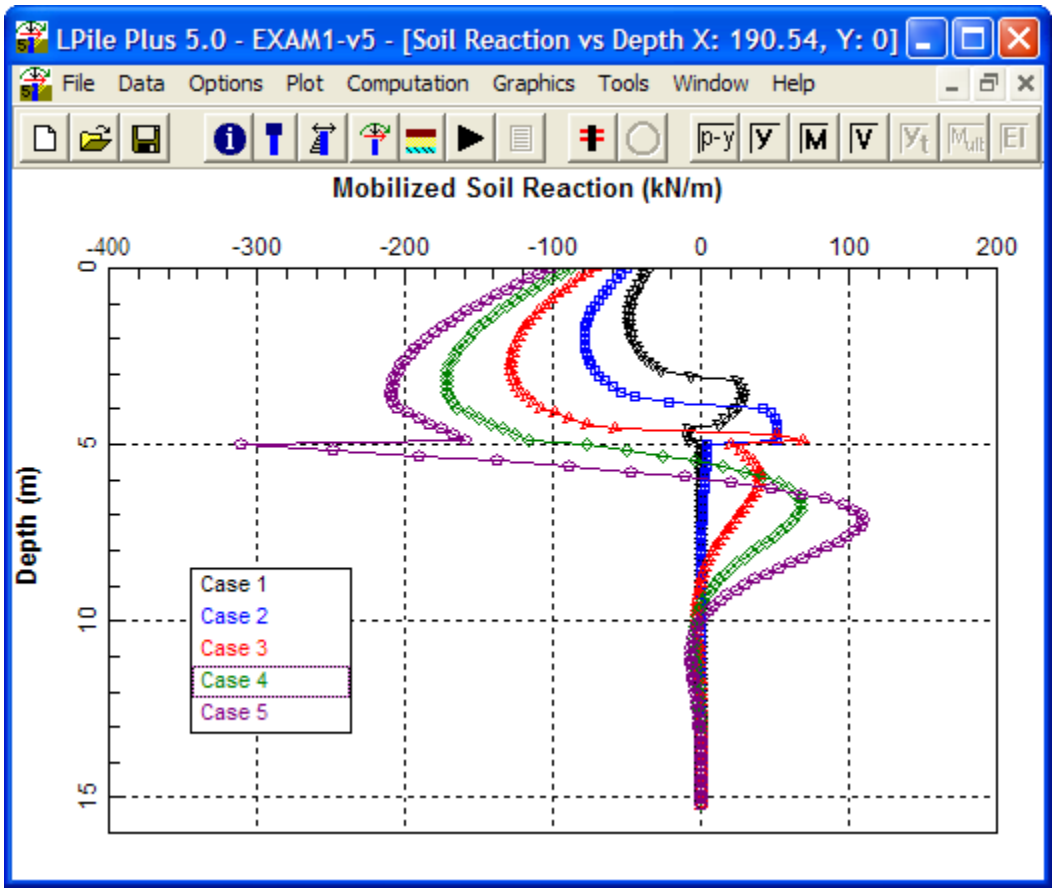


Shear Force vs. Depth



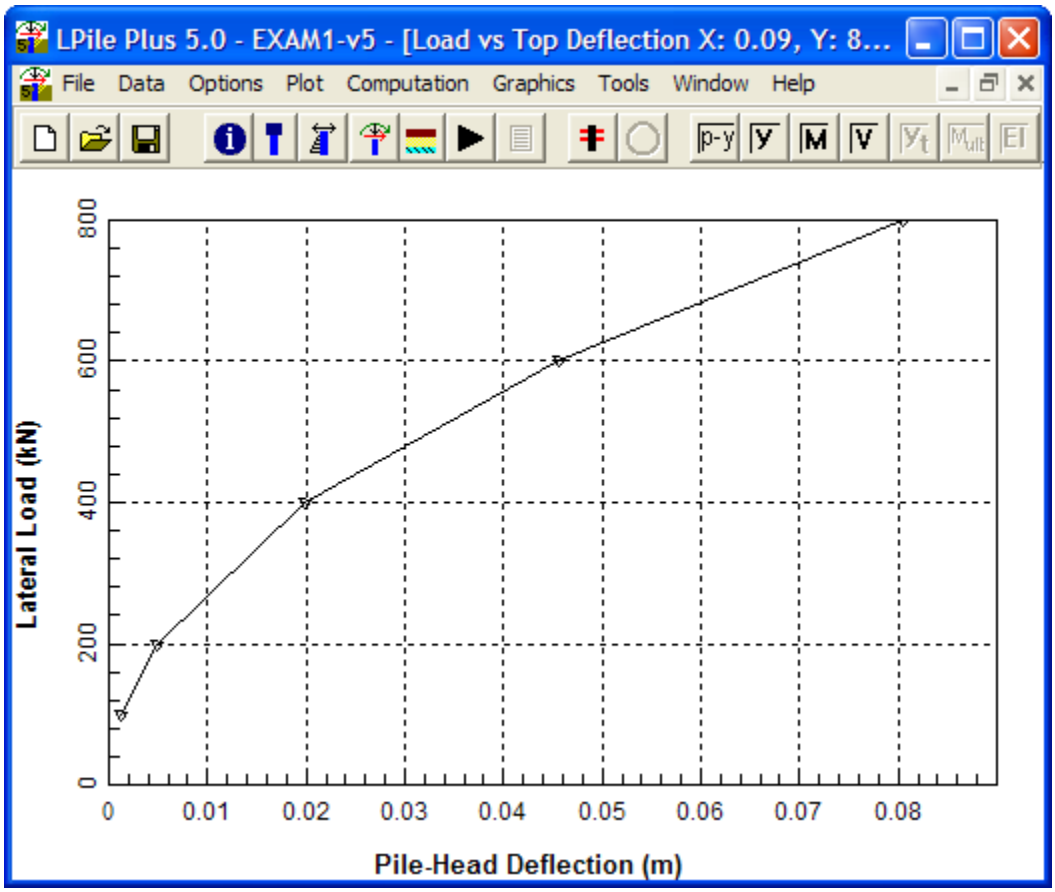
The user may select this command to display the curve of shear force versus depth. This curve is automatically generated in all analyses of a laterally-loaded pile. The number of points on the shear curve is equal to the selected number of pile increments. Several curves may be contained in this graphics if the user selects to input several load cases. When multiple curves are displayed, the display of individual curves can be triggered on or off by clicking on the color-coded label in the legend for the graph.

Soil Reaction vs. Depth



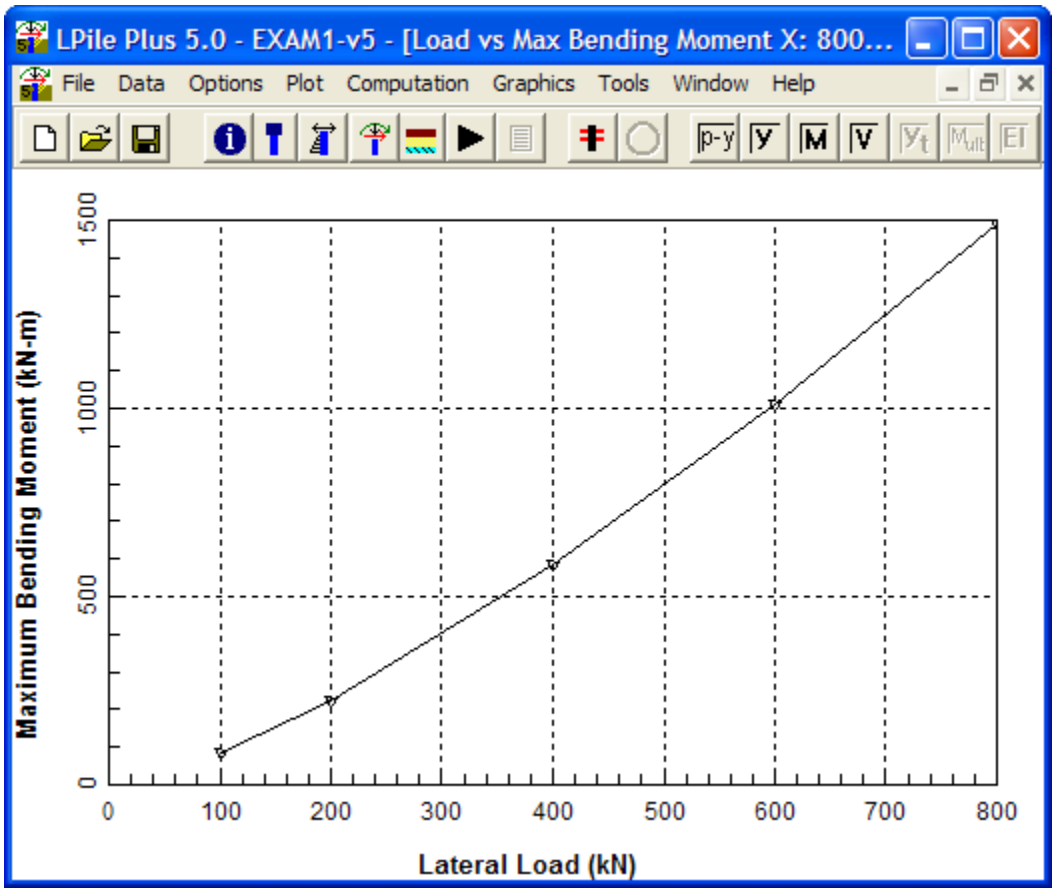
The user may select this command option to observe a graphical representation of the curve of soil reaction versus depth for the modeled pile. This curve is automatically generated in all analytical runs of a laterally loaded pile. The number of points on the soil-reaction curve is equal to the selected number of pile increments. A sample curve of the Soil Reaction vs. Depth command option is shown in the user's manual. Several curves may be contained in this graphics if the user selects to input several load cases. When multiple curves are displayed, the display of individual curves can be triggered on or off by clicking on the color-coded label in the legend for the graph.

Lateral Load vs. Pile-Head Deflection



This graph option is enabled only if the user specifies two or more load cases in the input data. The specified load cases must have varying lateral loads. The user may select this graph option to observe a graphical representation of the curves of applied lateral load versus pile-top deflection.

Lateral Load vs. Maximum Moment



This option provides a plot of maximum values of bending moments found anywhere in the pile length versus incremental lateral loads (applied at the pile head). This option is only enabled when the user specifies several load conditions with varying values of shear (lateral load) while keeping other loads constant.

Pile Length vs. Top Deflection

This option provides a plot of incremental pile lengths versus deflection at the pile head. This option is only enabled when Generate Pile Length vs. Top Deflection is check marked in the input (under the Options menu).

Lateral Load vs. K_{22}

This graphics option only becomes available if the user asked the program to generate foundation stiffness placing a check mark in Generate Foundation Stiffness under the Options menu.

This option provides a plot of lateral loads (applied at the pile head) versus K_{22}, where K_{22} represents K_{yy} in the 3x3 stiffness matrix that may be used to represent the modeled pile in superstructure models. This is only available when Generate Foundation Stiffness is specified in the input (under the Options menu).

Lateral Load vs. K_{23}

This graph is available if the user asked the program to generate foundation stiffness placing a check mark in Generate Foundation Stiffness under the Options menu.

This option provides a plot of lateral loads (applied at the pile head) versus K_{23}, where K_{23} represents $K_{y\theta}$ in the 3x3 stiffness matrix that may be used to represent the modeled pile in superstructure models. This is only available when Generate Foundation Stiffness is specified in the input (under the Options menu).

Moment vs. K_{32}

This graphics option only becomes available if the user asked the program to generate foundation stiffness placing a check mark in Generate Foundation Stiffness under the Options menu.

This option provides a plot of lateral loads (applied at the pile head) versus K_{32} where K_{32} represents $K_{\theta y}$ in the 3x3 stiffness matrix that may be used to represent the modeled pile in superstructure models. This is only available when Generate Foundation Stiffness is specified in the input (under the Options menu).

Moment vs. K_{33}

This graphics option only becomes available if the user asked the program to generate foundation stiffness placing a check mark in Generate Foundation Stiffness under the Options menu.

When available, the selection of this command option will show a curve of the K_{32} (sometimes written $K_{\theta\theta}$) component of a 3x3 foundation stiffness matrix.

K_{22} vs. Pile Top Deflection

This graphics option only becomes available if the user asked the program to generate foundation stiffness placing a check mark in Generate Foundation Stiffness under the Options menu.

This option provides a plot of pile top deflection versus K_{22}, where K_{22} represents K_{yy} in the 3x3 stiffness matrix that may be used to represent the modeled pile in superstructure models. This is only available when Generate Foundation Stiffness is specified in the input (under the Options menu).

K_{23} vs. Pile Top Rotation

This graphics option only becomes available if the user asked the program to generate foundation stiffness placing a check mark in Generate Foundation Stiffness under the Options menu.

This option provides a plot of pile top rotation versus K_{23}, where K_{23} represents $K_{y\theta}$ in the 3x3 stiffness matrix that may be used to represent the modeled pile in superstructure models. This is only available when Generate Foundation Stiffness is specified in the input (under the Options menu).

K_{32} vs. Pile Top Deflection

This graphics option only becomes available if the user asked the program to generate foundation stiffness placing a check mark in Generate Foundation Stiffness under the Options menu.

This option provides a plot of pile top deflection versus K_{32}, where K_{32} represents $K_{\theta y}$ in the 3x3 stiffness matrix that may be used to represent the modeled pile in superstructure models. This is only available when Generate Foundation Stiffness is selected under the Options menu.

K_{33} vs. Pile Top Rotation

This graphics option only becomes available if the user asked the program to generate foundation stiffness placing a check mark in Generate Foundation Stiffness under the Options menu.

When available, the selection of this command option will show a plot of pile top rotation versus K_{32} ($K_{\theta\theta}$) in the 3x3 stiffness matrix that may be used to represent the modeled pile in superstructure models.

Frequently Asked Questions

1. Is there any possibility to run LPile Plus Student Edition for Windows in batch mode?

It is not possible to run LPILE Plus Student Edition in batch mode due to the design of the program. The computations are divided between both the main Windows program and a dynamic link library module. Since the Windows program cannot be run in batch mode and the Windows program loads the data to and from the dynamic link library, LPILE cannot run independently from the Windows program.

2. According to software manuals: "k" is the gradient of the initial soil modulus versus depth, $E_{si} = E_{py}$, for the p-y-curves. In some terminologies, the soil stiffness is usually described by use of an equivalent Young's modulus, E_{Young}. What is the relationship between the "k"-value and Young's modulus, E_{Young}, of the soil? Is it possible to link these two parameters directly?

Equation 3.3, in Page 3-14 of the Technical Manual of LPILE Plus Version 5.0 provides a link between these two parameters ($E_{si} = K_{py} * X$ and E_{Young}). However, that equation is only for linear-elastic materials under plain-strain conditions. It is difficult to derive the direct connection between these two parameters because the E_{si} and K_{py} are dimension-dependent properties and E_{Young} is not.

3. If the strength of soil layers is only described by use of Young´s modulus, E_{Young}, what are the values of k that you would recommend for those soils?

We recommend use of the values for K_{py} that are listed in our technical manuals because these values were derived from results of field tests on various sizes of piles. The beam-column model used in LPile assumes the soil reaction as a series of lateral springs, and the *p-y* criteria are semi-empirical methods based on the simplification of

the soil behavior. Because of the nonlinear properties of soil at large strain, we use the strength parameters such as *c* (cohesive strength) and ϕ (internal friction angle) for the parameters to develop the full range of *p-y* curves (nonlinear spring reactions). We can not provide any *k* values if we do not have sufficient parameters of soil strength. The Young's modulus of the soil is not a direct parameter used by our program. Young's modulus of the soil may be used for pure linear-elastic analyses recommended in Poulos' method.

4. I have a question with regards to the selection of the k_{rm} parameter. The manual states that the value should be in the range of 0.0005 to 0.00005. However, the manual provides no indication on how to select this parameter based on laboratory or field testing. Is it a function of unconfined compressive strength, RQD, Elastic Moduli, or some other parameter?

We recommend that you examine the stress-strain curve of the rock sample. Typically, the k_{rm} is taken as the strain at 50% of the maximum strength of the core sample. Because limited experimental data are available for weak rock during the derivation of the *p-y* criteria, the k_{rm} from a particular site may not be in the range between 0.0005 and 0.00005. For such cases, you may use the upper bound value (0.0005) to get a larger value of y_{rm}, which in turn will provide a more conservative result.

5. I am interested in modeling drilled shafts supporting bridge piers which are subject to temperature induced lateral loads. I feel that this represents a cyclic-loading condition. However, in modeling this, should I select both the cyclic loading option as well as entering reduced soil stiffness parameters, or does the cyclic loading option automatically modify the soil parameters from what is entered?

The cyclic loading criteria for *p-y* curves was developed for low frequency cyclic loading due to sea waves. Typically this type of loading involves some degree of two-way loading, which causes greater degradation of load-transfer than does one-way loading. We do not recommend the use of the cyclic loading criteria for *p-y* curves to be applied to the thermal loading problem because the period of thermal loading may be 24 hours or longer, not the 10 to 12 seconds for which the cyclic loading criteria were developed. Thus, we recommend use of the static loading criteria for this problem.

What is more important is to examine the magnitude of the range of displacements resulting from thermal loading. Typical coefficients of thermal expansion of concrete is in the range of 0.000006/deg. F. It would have to be a large structure before these types of deformations would become important. However, we would recommend the application of a prescribed displacement in place of a loading force when analyzing this type of problem.

6. If I specify 0.25 in. as excessive deflection under Control Options I get the same lateral force required to obtain that deflection as I do when using the displacement-and-moment boundary condition. What is wrong with this procedure?

This is not the purpose of the excessive deflection control option. This option is used to stop the program before an overrun/underrun execution error occurs and to prevent the need to reboot the computer. The pile-head boundary condition is used to set deflection requirements.

7. If I use the displacement-slope pile head (Type 5) condition, the shear and moment diagrams sometimes, but not always, have large spikes in the curves at the top two nodes on the pile. What is happening here?

This condition usually occurs when performing an analysis with the following conditions: (1) one is modeling a pile with the pile head fixed against rotation, (2) the analysis includes the effects of nonlinear bending (Type 3 Analysis), and (3) additional loading using the soil displacement option is being used. The spikes in the shear and moment diagrams are a result how the beam column differential equation is solved for the Type 5 pile head condition. In the solution of the differential equation for the pile head conditions Types 1 through 4, the solution obtains displacement values for two imaginary nodes above the pile head. The values of bending moment and shear force are computed using central finite difference expressions for the second and third derivatives of displacement versus depth. With the Type 5 pile head condition, the displacement values for the two imaginary nodes above the pile head are not obtained in the solution because a different differential equation solution algorithm is used. This prevents the use of central finite difference expressions for the evaluation of the shear force and moment values at the top two nodes. Instead, higher order forward difference expressions are used. These expressions use displacement values at the top 4 nodes for moment and top 5 nodes for shear force. Often when nonlinear bending effects are included for concrete foundations, the value of bending stiffness across these nodes may have a large variation due to cracking. The variation of bending stiffness may also be increased in cases involving soil displacements. All of these effects combine to cause the spikes in the shear and moment curves.

The following approach is recommended to eliminate the spikes in the shear and moment curves. Use the displacement-moment (Type 4) pile head condition and input a range of moments that result in a range of solutions that bracket the desired pile head rotation value. Then, determine the necessary value of pile head moment that results in the desired pile head rotation by interpolation between the trial values. Next, reanalyze the pile when applying the interpolated moment value. Repeat the process until the desired level of fit is achieved. The plots of the shear force and moment diagrams for this solution should not have the spikes at the top two nodes found before.

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